

LSE - FCA CAPSTONE REPORT

# IS BRITAIN'S FINANCIAL SECTOR

# TOO LARGE?



**MARCH 2025**

Prepared by



School of  
Public Policy

Prepared for



FINANCIAL  
CONDUCT  
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Prepared by: School of Public Policy,  
The London School of Economics and Political Science

Prepared for: The Financial Conduct Authority

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Word count: 14,971

## Disclaimer

This research was undertaken as part of the Master of Public Administration (MPA) Capstone Project at the London School of Economics and Political Science (LSE) School of Public Policy (SPP), in collaboration with the Financial Conduct Authority (FCA). The authors were not employed by the FCA, this project was conducted solely as part of the MPA curriculum. While the findings may contribute to the FCA's work by providing valuable insights and fostering stimulating debate, the views expressed herein are solely those of the authors and do not reflect the positions of the FCA or the LSE SPP. Any errors or omissions in this research are the responsibility of the individual authors and should not be attributed to the FCA or the LSE SPP.

## Acknowledgments

We extend our heartfelt gratitude to Luke Pate, Eric Morrison, Alex Stroud, Connor O'Loughnan, and Haris Irshad from the Financial Conduct Authority (FCA) for their invaluable guidance, availability and support throughout this project. Their continuous feedback has been immensely beneficial, and we are deeply appreciative of their contributions. It has been an honour for us to work with the FCA. We also wish to thank Professor Frank Muci-Lander at the LSE School of Public Policy for his insightful mentoring. This Capstone Project has profoundly shaped our graduate school experience, and we are fortunate to have had such enriching support.

*Note. Cover page image: 'Buildings With Glass Windows', by Expect Best, 2017, photograph, Pexels (free stock photo). <https://www.pexels.com/photo/buildings-with-glass-windows-351264/>*

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## Acronyms and abbreviations

| Acronym/Abbreviation | Definition                                             |
|----------------------|--------------------------------------------------------|
| BoE                  | Bank of England                                        |
| CMA                  | Competition and Markets Authority                      |
| EPRS                 | European Parliamentary Research Service                |
| FCA                  | Financial Conduct Authority                            |
| FCDO                 | Foreign, Commonwealth & Development Office             |
| FDI                  | Foreign Direct Investment                              |
| FOS                  | Financial Ombudsman Service                            |
| FSD                  | Financial Structure Database                           |
| FSS                  | Financial Services Sector                              |
| GDP                  | Gross Domestic Product                                 |
| GFC                  | Global Financial Crisis                                |
| GFCF                 | Gross Fixed Capital Formation                          |
| GFDD                 | Global Financial Development Database                  |
| GGs                  | Growth Guarantee Scheme                                |
| GVA                  | Gross Value Added                                      |
| HM                   | His Majesty's                                          |
| HSBC                 | Hongkong and Shanghai Banking Corporation              |
| ILO                  | International Labour Organization                      |
| ILOEST               | ILO Modelled Estimates                                 |
| ILOSDG               | ILO SDG Labour Market Indicators Database              |
| LCF                  | London Capital & Finance                               |
| NBER                 | National Bureau of Economic Research                   |
| OECD                 | Organisation for Economic Co-operation and Development |
| OLS                  | Ordinary Least Squares                                 |
| ONS                  | Office for National Statistics                         |
| PRA                  | Prudential Regulation Authority                        |
| PSR                  | Payment Systems Regulator                              |
| R&D                  | Research & Development                                 |
| SDG                  | Sustainable Development Goals                          |
| SEC                  | Securities and Exchange Commission                     |
| SFO                  | Serious Fraud Office                                   |
| SME                  | Small and Medium Enterprise                            |
| STAN                 | OECD Structural Analysis Industry Database             |
| SVB                  | Silicon Valley Bank                                    |
| TTC                  | Total Tax Contribution                                 |
| UK                   | United Kingdom                                         |
| UN                   | United Nations                                         |
| US                   | United States of America                               |
| WDI                  | World Bank World Development Indicators                |
| WGI                  | World Bank World Governance Indicators                 |

## Executive summary

The UK's financial sector is a cornerstone of its economy, providing employment, and tax revenues, and acting as a vital intermediary for investment and capital allocation. In 2024, it accounted for nearly 9% of national GDP (ONS, 2025b).

While the economic literature has long established a positive relationship between financial sector development and economic growth, concerns have grown over whether continued expansion remains beneficial – particularly given the UK's stagnant productivity and output growth since the Global Financial Crisis (GFC). Several researchers have proposed an inverted U-shaped relationship between finance and growth, suggesting that financial sector expansion negatively affects economic growth once domestic credit to the private sector reaches 90-100% of GDP (Arcand et al., 2012; Cecchetti & Kharroubi, 2012).

However, these studies do not claim causality, and the finance-growth relationship appears highly sensitive to factors such as income levels, institutional context, and the nature of credit. Furthermore, much of the empirical basis stems from countries most affected by the GFC, with limited post-2009 evidence.

This report investigates whether there is a growth-maximising threshold to the size of a country's financial sector by analysing a panel dataset that covers a cross-country sample from 1960 to 2024. This study critically engages with Cecchetti & Kharroubi's (2012) methodological framework, replicating and extending their methodological design to test the robustness of their conclusions using more recent and comprehensive data.

First, this study conducts a panel regression of GDP per capita growth on private credit to GDP, financial employment share, and their squared terms, both for the full country sample and for high-income countries. Second, it addresses methodological limitations by incorporating country and time-fixed effects, clustering standard errors, refining control variables, and testing the assumptions of a quadratic specification using non-parametric LOWESS curves.

The main finding is that there is no evidence of a non-linear relationship between financial development and economic growth, reflecting the complex, context-specific, and time-sensitive nature of this interaction. By replicating and subsequently refining the original

methodology, this study reveals that the relationship lacks statistical robustness, with results proving highly sensitive to model design and variable selection.

A key methodological concern is the use of correlational models to capture the endogenous relationship between growth and financial depth. Additionally, the widespread use of second-degree polynomial specifications in existing literature imposes excessive rigidity on the analysis. As a result, the findings should be interpreted with caution, as the evidence does not support a direct or consistent causal relationship between finance and growth.

Importantly, the study challenges the notion of an inverted U-shaped relationship between financial development and economic growth, along with the associated idea of a definitive growth-maximising threshold for financial sector expansion.

Focusing on the UK, this analysis does reveal an apparent inverted U-shape in the residuals. However, this pattern is likely reflective of the long-run impacts of the GFC and broader economic stagnation rather than the financial sector's size. This reinforces the central argument that country-specific and time-specific factors are essential in interpreting the finance-growth relationship.

Accordingly, this report recommends measures to enhance the financial sector's contribution to the UK economy, rather than seeking to reduce its size through regulation or exposing it to greater risks through poor deregulation. First, improving SME access to finance through an expanded Growth Guarantee Scheme and reduced reliance on personal guarantees could boost innovation and job creation. Second, enhancing coordination among existing regulators would improve oversight without straining capacity. This could be achieved through joint supervision and enforcement initiatives, shared data systems for risk monitoring, and the creation of a formal coordination council. Finally, encouraging investment beyond London through better infrastructure and R&D support could help unlock the sector's potential to drive regional growth.

## 1. Introduction

In 1986, Margaret Thatcher's 'Big Bang' deregulated the London Stock Exchange, transforming the City of London into one of the world's leading financial hubs. In the following decades, Britain's financial markets have grown substantially. The financial services sector (FSS) now contributes nearly 9% of GDP, while manufacturing shares have declined and regional inequalities have widened (Stansbury et al., 2023). The sector provides significant employment, tax revenues, and a major share of exports. Beyond its direct economic contributions, financial services play a key role in facilitating investment, managing risk, and ensuring efficient capital allocation (Morrison et al., 2024).

Historically, research has supported a positive link between financial development and growth. Despite this, concerns persist over whether the continued expansion of the FSS remains beneficial. Some experts have argued that further growth could divert resources away from more productive industries and increase systemic risk, ultimately proving counterproductive for long-term growth (Stewart, 2024). These concerns are echoed in empirical studies linking large financial sectors to heightened economic vulnerability, particularly in the aftermath of the GFC (Grjebine & Tripier, 2017; Mathonnat, 2017). Some studies suggest that this relationship may be non-linear, with diminishing or even negative returns beyond a certain threshold (Arcand et al., 2012; Cecchetti & Kharroubi, 2012). Yet, significant uncertainties remain: estimated thresholds vary widely, and the causal direction of the relationship is still debated. Much of the existing evidence relies on correlations, and the finance-growth nexus appears highly context-dependent, influenced by income levels, financial structures, and the type of credit provided.

Additionally, the regulatory landscape has largely evolved significantly since the GFC. Reforms such as stricter capital and liquidity requirements, enhanced supervision, and clearer resolution mechanisms have focused on improving resilience and reducing systemic risks. The Bank of England has emphasised that the robustness of the banking system is more important than its size (Bush et al., 2014), suggesting that the traditional link between financial sector size and economic performance may have changed and warrants re-examination.

Yet, the UK continues to face a 'growth puzzle': productivity has stalled, and real GDP per capita has barely increased since the GFC. While the financial sector plays a central role in the

country's service-based economy, its long-term sustainability and contribution to growth are increasingly being questioned. This raises a critical policy question: does the continued expansion of the financial sector enhance economic productivity, or has it reached a point where further growth hinders broader economic performance? In other words, is the UK's financial sector too large?

This study aims to assess whether the UK FSS has grown beyond its optimal size. Specifically, it examines whether there is an empirical threshold beyond which financial sector growth ceases to support or begins to harm economic growth. To this end, it pursues three secondary objectives:

1. Evaluate the financial sector's contribution to the UK economy through investment, productivity, employment, tax revenues, and international trade.
2. Replicate and refine existing empirical models – particularly the inverted U-shaped relationship proposed by Cecchetti & Kharroubi (2012) – to test the robustness of the 'Too Much Finance' hypothesis.
3. Provide evidence-based policy recommendations to ensure the financial sector supports broad-based and sustainable economic growth.

Debates around the optimal size and role of the FSS have intensified in recent years, particularly in the context of Brexit and shifting economic policymaking. Understanding whether the sector's current trajectory supports economic resilience or introduces new vulnerabilities is essential to inform effective policy.

To address these gaps, this study draws on updated data and an extended set of countries to evaluate the validity of the 'Too Much Finance' hypothesis and explore whether the UK's financial sector has grown beyond a level consistent with long-term economic performance. Beyond contributing to theoretical debates, the findings offer timely insights for UK policymakers, particularly given the FCA's secondary objective to promote international competitiveness and growth. Ultimately, this research supports the development of policy and regulation that balances financial sector expansion with long-term economic resilience.

The rest of this study is structured as follows. Section 2 assesses the concern that the UK's financial sector may have grown too large, potentially undermining broader economic growth,

and outlines the key challenges associated with its size. Section 3 provides a review of the existing literature on the finance-growth relationship, establishing the conceptual framework used in this study. Building on this, Section 4 examines Britain's FSS' contribution to the economy, drawing connections to broader theoretical debates.

In the second half of the report, Section 5 outlines the research hypotheses, design, and methodology, drawing on Cecchetti & Kharroubi (2012). Section 6 presents the study's findings, further discussed in Section 7. Section 8 offers policy recommendations to ensure the financial sector positively contributes to long-term economic growth. Finally, Section 9 concludes and summarises key insights.

## 2. Policy problem and research question

The financial sector is one of the largest industries in the UK, representing approximately 9% of total GVA in 2022. While it plays a crucial role in facilitating investment, attracting a skilled workforce, and generating substantial tax revenue, concerns persist around whether an excessively large financial sector may hinder the growth of other industries. This raises the following research question: *‘Does Britain’s financial sector’s continued expansion enhance economic productivity, or has it reached a stage where further growth hinders broader economic performance? In other words, is the UK’s financial sector too large?’*

Concerns about the FSS’ size and influence have grown, particularly after the GFC, which rapidly triggered a global recession (Szczepanski, 2019). Given the financial sector’s deep interconnections with the broader economy, disruptions in financial markets entail far-reaching and lasting consequences.

### 2.1. The link between financial sector size, economic volatility and systemic risks

Empirical research shows that economies with larger financial sectors are more prone to crises and experience greater economic contractions during downturns (Mathonnat, 2017). While a well-developed financial sector can drive economic growth, its expansion also amplifies systemic risks, making crises more frequent and severe (Grjebine & Tripier, 2017).

This phenomenon was seen in the aftermath of the GFC. Before the crisis, the UK experienced solid growth, with GDP expanding at an average rate of 2.7% and the financial sector accounting for 9.4% of total output. After the crisis, GDP contracted sharply by 4.8%. The US, with a financial sector of around 7% of GDP, also saw a sharp 4.3% contraction. In contrast, France, with a financial sector roughly half the size, experienced a significantly milder contraction of only 2.9% (OECD, 2025a).

Despite the regulatory reforms introduced following the GFC, the financial sector continues to exhibit vulnerabilities and behaviours that could contribute to future crises, such as attempts to bypass controls for profit maximisation for instance. While post-crisis regulations have focused on FSS day-to-day operations (micro-prudential measures), they have overlooked the evidence that major financial crises are often systemic in nature (Danielsson & Uthemann, 2023).

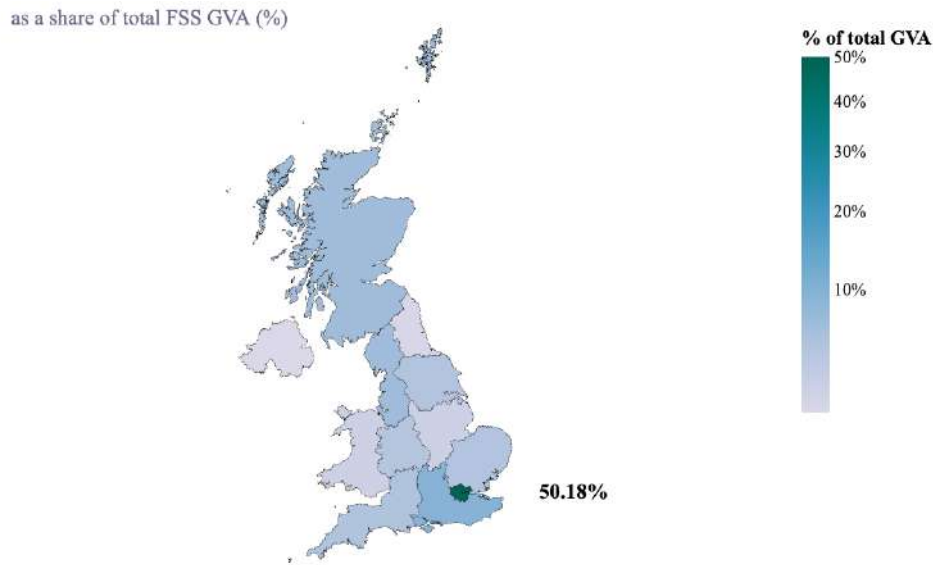
A key regulatory response to the GFC has been the implementation of the Basel III agreement that puts forward capital requirements and more efficient risk management for banks. In the UK, the aftermaths of the crisis have led the government to implement the Banking Act and the Banking Reform Act between 2009 and 2013, separating retail and investment banking to prevent banks from exposing client deposits to excessive risk (Javid & HM Treasury, 2013). These acts have also eased the rules for failing or insolvent banks to be acquired by private actors. For instance, following the collapse of Silicon Valley Bank (SVB) in 2023, authorities facilitated the purchase of SVB UK by HSBC for £1 (HSBC, 2023). These mechanisms strengthened macroprudential oversight of the financial system.

However, banking supervision alone cannot eliminate systemic risks. A systemic crisis triggered by legally authorised high-risk investments can still entail severe consequences, particularly in heavily FSS-dependent economies. While stricter regulations are necessary to mitigate systemic risks, they may limit financial institutions' profitability by restricting certain types of investments. In a global landscape where financial centres compete for capital and talent, concerns about the UK's competitiveness under an overly restrictive regulatory regime are therefore not unfounded.

These dynamics highlight the ongoing trade-off between financial stability and competitiveness. As the financial sector evolves, so do the risks and structural vulnerabilities that may not be fully addressed by current regulations. Policymakers must strike a balance between ensuring resilience to shocks and preserving the sector's capacity to support growth and innovation. Understanding the link between financial sector size, systemic risk, and volatility is crucial for designing both effective and proportionate regulations.

## **2.2. The broader economic and social implications of financialisation**

Beyond financial stability concerns, the gains from the financial industry (measured in gross value added) are highly unequally distributed across the UK. London concentrates most of the financial added value in the country: 48% in 1998 and 50% in 2022, followed by the South East at under 10% of total GVA (Figures 2.1-2.2) (ONS, 2024b). The contribution of all other regions to total FSS GVA is noticeably small. In total, financial services represent 20% of Greater London's total economic output (ONS, 2024b).

**Figure 2.1.** Finance and insurance share of total GVA (%) by UK ITL1 regions, 2022

Source: authors' elaboration using ONS Blue Book 2024 data.

**Figure 2.2.** Finance and insurance GVA (real GBP billions, 2019 money value) by UK ITL1 regions, 1998 (left) vs 2022 (right)

Source: authors' elaboration using ONS Blue Book 2024 data.

This uneven distribution of FSS activity reinforces existing regional disparities in economic performance and opportunity. The concentration of high-value financial services in London not only limits the sector's contribution to broader national growth, but also deepens geographic inequalities in income, investment, and employment. As a result, efforts to better understand and manage financialisation should not only consider its impact on stability and efficiency but also its role in shaping Britain's wider economic geography.

### 3. Literature review

#### 3.1. The traditional finance-growth nexus

The impact of finance on economic growth has long been studied in economics. In the 1960s, the National Bureau of Economic Research (NBER) assessed the relationship between overall financial development and economic activity in 35 countries using pre-1964 data (Goldsmith, 1959). This study found a positive correlation between financial development and economic growth, though it did not draw causal interpretations.

Subsequently, a large and expanding body of literature has examined the effects of financial development on economic growth, attempting to establish causality using a range of econometric techniques, including cross-country, time series, panel data, and firm-level studies (Beck et al., 2005; Jalil & Ma, 2008; Khan, 2003; King & Levine, 1993). While much of this research initially supported the view that financial development fosters economic growth, more recent studies have yielded mixed results, leading to an emerging debate on whether excessive financialisation could, instead, hinder economic progress.

One of the most widely cited studies analysed over 80 countries from 1960 to 1989 and found a strong positive association between financial sector development and real per capita GDP growth (King & Levine, 1993). They argued that financial development enhances growth by improving resource allocation, mobilising savings, strengthening corporate governance, and enabling risk-sharing.

This perspective was further reinforced by Odedokun (1996), who examined 71 countries and found that financial intermediation promoted economic growth in 85% of cases. Similarly, recent studies found a positive finance-growth relationship across selected European Union, African, and emerging market economies between 1980 and 2010 (Adusei, 2013; Leitão, 2010). Using more recent data, Guru & Yadav (2019) examined BRICS economies (Brazil, Russia, India, China, and South Africa) from 1993 to 2014 and found that banking development promotes growth when paired with stock market efficiency.

Extensive evidence also suggests that, in the short run, financial development entails a trade-off between higher economic growth and increased volatility, including a greater risk of crises, particularly in high-income countries (Beck et al., 2014; Loayza et al., 2017).

### 3.2. The ‘Too Much Finance’ hypothesis

Following the GFC, researchers increasingly questioned whether financial sector overgrowth could derail economic progress rather than foster it. This debate led to the emergence of the ‘Too Much Finance’ hypothesis, which posits that financial expansion initially boosts growth before hindering economic performance beyond a certain threshold (Epstein & Crotty, 2013; Gründler & Weitzel, 2012; Mamonov et al., 2018; Samargandi et al., 2015; Santomero & Seater, 2000).

These empirical studies challenge the previous linear perspective, arguing that the relationship between finance and growth is neither uniform nor constant over time. They identify an inverted U-shaped relationship between financial sector development and economic growth using data from both advanced and emerging economies from 1980 to 2014. While some studies have explored a causal relationship between financial development and growth, it is important to note that papers identifying an inverted U-shaped association between finance and growth do not necessarily argue for causality.

Using domestic credit to the private sector as a proportion of GDP to measure financial development, several studies estimate the turning point – where credit starts to negatively impact growth – to be around 90-100% of GDP (Arcand et al., 2012; Cecchetti & Kharroubi, 2012; Law & Singh, 2014). However, using the same measure, a different study estimates a higher threshold of 142% of GDP (Swamy & Dharani, 2019).

Similar threshold effects are found using other measures of the financial development. Cecchetti and Kharroubi (2012) find an inverted U-curve between financial sector employment and GDP growth, where FSS employment above 3.5% of the workforce detrimentally impacts growth. Using data from 63 countries (1980–2014), Mamonov et al. (2018) find that different financial development indicators have distinct optimal thresholds: 84% of GDP (private sector loan market), 32% of GDP (domestic corporate bond market), and 23% of GDP (external long-term corporate debt market).

Besides economic growth, Easterly et al. (2000) find a U-shaped relationship between output volatility and financial sector depth, where volatility rises once private credit surpasses 100% of GDP. While financial sector development and a well-established financial system initially reduce growth volatility, the risks rise as the sector expands. Ultimately, stability may decline when the financial system becomes excessively large.

Although the exact threshold varies, these findings repeatedly suggest the existence of a non-linear relationship. Arcand et al. (2012) claim that their results align with the ‘vanishing effect’ of finance, meaning that the benefits of financial development eventually disappear as the FSS expands. However, the validity of their findings remains open to discussion, despite their assertion that output volatility, banking crises, institutional weaknesses, or regulatory differences do not influence results.

This reverse U-shaped relationship is often attributed to the so-called ‘Financial Dutch Disease’, which suggests that countries with oversized financial sectors may face economic challenges similar to those observed in commodity-rich economies suffering from the resource curse (Christensen et al., 2016; Shaxson & Christensen, 2013). Originally introduced in the 1970s to describe the adverse effects of natural resource booms on the Dutch manufacturing sector, ‘Dutch Disease’ occurs when large capital inflows lead to currency appreciation, reducing the competitiveness of tradable sectors (Frankel, 2010; Reisinezhad, 2020).

Botta (2015) extends this concept to the financial sector, arguing that a disproportionate financial sector can purportedly distort the economy by driving large foreign direct investment (FDI) inflows into financial markets, triggering currency appreciation. The resulting exchange rate pressures are shown to interact with capital movements in a self-reinforcing cycle that ultimately weakens long-term manufacturing development. Over time, this persistent appreciation is claimed to lead to declining economic and labour productivity growth.

As the FSS expands, it is not only argued to absorb a disproportionate share of capital resources, but also talent. High salaries in finance are believed to attract skilled professionals away from government, research, and innovation-driven industries, weakening sectors that are essential for long-term economic growth (Cecchetti & Kharroubi, 2012, 2015). This reallocation of

human capital is often said to create a self-reinforcing cycle, where financial activities benefit financial development at the expense of broader economic development.

Additionally, Dabla-Norris & Narapong (2013) argue that large financial landscapes heighten consumption and investment volatility, potentially increasing long-term economic instability. The cyclical nature of finance-driven growth, marked by boom-and-bust cycles, is believed to amplify macroeconomic vulnerabilities and crisis risks, an argument that gained prominence particularly after the GFC.

### **3.3. Potential concerns with these models**

Nevertheless, existing research that models the relationship between financial development and economic performance using a second-degree polynomial has a critical limitation: the inability to establish a causal connection between these two variables. While these studies suggest that financial expansion beyond a certain threshold hinders economic growth, this assertion lacks solid empirical support. As discussed further in this document, these models suffer from endogeneity, as various external factors could simultaneously influence both financial development and economic performance.

Additionally, omitted variables or reverse causality may be responsible for the observed pattern, as economic growth itself may drive changes in FSS size. Indeed, researchers have divided views on the relationship between credit to the private sector and economic growth. Some argue for a unidirectional causal flow from economic growth to domestic credit (Omri et al., 2015), while others identify a bidirectional relationship between bank credit and economic growth (Raz, 2013). Studying a sample of 40 developing countries, Mhadhbi et al. (2020) find that the direction of the causal link between financial development and economic growth is highly dependent on country-specific characteristics.

These methodological flaws can potentially introduce bias, making the identified association correlational rather than causal. As a result, the findings of these studies should be interpreted cautiously, as they do not provide conclusive evidence of a direct causal effect.

Furthermore, the studies reviewed critically rely on data from a limited number of observations to identify a turning point in the relationship, primarily data from countries severely impacted

by the GFC. This suggests that the observed pattern may be highly sensitive to their inclusion, raising concerns that omitting them could render the relationship statistically insignificant or void. Notably, Abdul Bahri et al. (2018) analyse the post-2009 period and find that the nonlinear relationship between financial development and economic growth follows a U-shape, contradicting the ‘Too Much Finance’ literature. This means financial development appears to accelerate economic growth past a certain threshold.

Expanding the sample size and analysing a panel of approximately 100 countries from the mid-1990s to 2012, Botev et al. (2019) find no evidence supporting the inverted U-shaped relationship between finance and economic performance, further suggesting that the ‘Too Much Finance’ framework may not hold.

### **3.4. The nuances of the finance-growth relationship**

Besides the inconclusiveness of findings related to the ‘Too Much Finance’ hypothesis, it is important to note their lack of universality. These results are highly context-dependent and require nuanced interpretation, as they vary according to factors such as country income levels, time periods, financial structures, and measurement approaches.

#### **3.4.1. The importance of time periods**

Several studies suggest that the finance-growth relationship is not static over time. One school of research finds that while finance positively impacted growth between 1960 and 1989, this effect vanished post-1990 and became negative after 2000 (Gründler & Weitzel, 2012; Rousseau & Wachtel, 2009). Furthermore, the impact became significantly negative when estimating the effect with data up to 2010, suggesting that the post-GFC sharp decline in per capita income may explain this shift.

These findings support the argument that unregulated financial expansion can harm long-term productivity, providing a potential explanation for why highly financialised economies have experienced prolonged sluggish growth post-GFC. However, post-crisis financial regulations designed to curb excessive risk-taking in lending may have reshaped these dynamics, as evidenced by Abdul Bahri et al. (2018).

### 3.4.2. The role of income levels and development stages

There is a broad consensus that the relationship between the FSS and economic growth varies significantly with country-specific characteristics, particularly based on income levels and development stages. This cross-country heterogeneity challenges the validity of a single average estimate (Favara, 2003).

However, empirical studies do not fully agree on the direction of this effect. Odedokun (1996) finds that the growth-promoting effects of financial intermediation are stronger in low-income than high-income countries, implying that financial expansion is more beneficial in early-stage economic development. An explanation is that developed countries already have strong financial systems, thus additional financial development offers fewer marginal benefits.

Rioja & Valev (2004) test the differential effects of finance on growth in countries with three levels of financial development and find that finance positively impacts growth in middle and high-income countries, but its effect in low-income countries remains unclear.

In contrast, Maciejewski & Głodowska (2020) support the ‘Too Much Finance’ hypothesis but argue that the negative effects of financial sector expansion are primarily observed in transition and middle-income economies. Their findings suggest that financialisation does not have a statistically significant impact on economic growth in high-income economies.

Overall, while financial intermediation generally supports economic growth, its growth-promoting effects appear to be more pronounced in developing economies. In highly developed financial markets, where capital allocation mechanisms are already sophisticated, further expansion of the financial sector may have a limited impact on growth.

### 3.4.3. The importance of structural differences

Beyond time and country samples, some scholars argue that the type of credit provided by financial institutions matters. Beck et al. (2012) find that enterprise loans are positively correlated with growth, whereas household loans do not trigger any growth effects; they are often used for consumption rather than productivity-enhancing activities. This research argues that a financial sector dominated by household loans does not lead to significant income

growth, positing the need to distinguish between productive and non-productive credit expansion in future research. Some evidence also suggests that intermediation activities foster long-term growth and reduce volatility, whereas financial sector expansion in other activities has no long-term impact (Beck et al., 2014).

Table 3.1 displays a summary of the previously discussed literature and its key findings.

| <b>Table 3.1. Summary of existing literature: research methods and main findings</b> |                                                    |                                |                                                                                |                                                                                                                   |                                                              |
|--------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Authors                                                                              | Sample size                                        | Data & sample period           | Methodology                                                                    | Relationship between financial development and economic growth                                                    | Interpretations                                              |
| Positive Relationship                                                                |                                                    |                                |                                                                                |                                                                                                                   |                                                              |
| King and Levine (1993)                                                               | 80 economies                                       | Cross-sections (1960-1989)     | Ordinary Least Squares (OLS) and Instrumental Variables (IV) regressions       | +ve                                                                                                               | Correlation                                                  |
| Odedokun (1996)                                                                      | 71 developing economies (LDCs)                     | Time-series data (1960s–1980s) | OLS regressions and panel data estimations                                     | +ve (~85% of cases, stronger effect in low-income economies)                                                      | Correlation                                                  |
| Beck et al. (2005)                                                                   | 54 economies                                       | Firm-level data (1990s)        | Generalised Method of Moments (GMM) and panel regressions                      | +ve                                                                                                               | Correlation                                                  |
| Leitao (2010)                                                                        | 31 economies (Eu-27 and BRIC)                      | Panel data (1980–2006)         | GMM estimation                                                                 | +ve                                                                                                               | Correlation                                                  |
| Adusei (2013)                                                                        | 24 African economies                               | Panel data (1981–2010)         | GMM estimation                                                                 | +ve                                                                                                               | Bidirectional causality                                      |
| Guru and Yadav (2019)                                                                | BRICS (Brazil, Russia, India, China, South Africa) | Panel data (1993–2014)         | GMM estimation                                                                 | +ve                                                                                                               | Correlation                                                  |
| Mixed/ Context-Dependent Results                                                     |                                                    |                                |                                                                                |                                                                                                                   |                                                              |
| Deidda and Fattouh (2002)                                                            | 119 developed and developing economies             | Cross-sections (1960-1989)     | Threshold regression model                                                     | Mixed (+ve for high-income economies and insignificant for low-income economies)                                  | Correlation                                                  |
| Rioja and Valev (2003)                                                               | 74 economies                                       | Panel data (1961–1995)         | GMM dynamic panel estimation                                                   | Mixed (+ve for middle- and high-income economies and unclear for low-income economies)                            | Correlation                                                  |
| Jalil and Ma (2008)                                                                  | Pakistan, China                                    | Time-series data (1960–2005)   | Autoregressive Distributed Lag (ARDL) bounds testing approach to cointegration | Mixed (+ve for Pakistan but insignificant for China)                                                              | Correlation                                                  |
| Beck et al. (2012)                                                                   | 45 economies                                       | Panel data (1994–2005)         | GMM estimation                                                                 | Mixed (+ve for Enterprise credit, insignificant for household credit)                                             | Correlation                                                  |
| Fethi and Katircioglu (2015)                                                         | United Kingdom                                     | Time-series data (1970–2012)   | Seasonal cointegration analysis and vector error correction model (VECM)       | Mixed (Financial sector supports long-term growth, but stock market volatility negatively impacts GDP per worker) | Causality from finance to growth (supply-leading hypothesis) |

|                                             |                            |                                            |                                                                            |                                                                                                                     |             |
|---------------------------------------------|----------------------------|--------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------|
| Maciejewski and Głodowska (2020)            | 97 economies               | Panel data (2000–2017)                     | Linear ordering method (synthetic indicator) and simple panel regression   | Mixed (-ve for transition and middle-income economies, and insignificant for high-income economies)                 | Correlation |
| Non-linear Relationship / Threshold Effects |                            |                                            |                                                                            |                                                                                                                     |             |
| Easterly et al. (2000)                      | 60–74 economies            | Panel data (1960–1997)                     | Growth volatility analysis                                                 | Non-linear relationship between financial depth and the volatility of output growth                                 | Correlation |
| Rousseau and Wachtel (2011)                 | 84 economies               | Cross-sectional and panel data (1960–2004) | Cross-sectional and GMM dynamic panel estimation                           | Non-linear (Disappearance of positive effects across time)                                                          | Correlation |
| Cecchetti and Kharroubi (2012)              | 50 economies               | Panel data (1980–2009)                     | GMM and threshold regression                                               | Inverted U-shaped (GDP growth declines when financial sector employment exceeds 3.5% of total employment)           | Correlation |
| Arcand et al. (2012)                        | 63–97 economies            | Cross-sections and panel data (1960–2010)  | Dynamic panel threshold regressions                                        | Inverted U-shaped (Identifies a finance-growth threshold at 100% of GDP)                                            | Correlation |
| Gründler and Weitzel (2013)                 | 188 economies              | Panel data (1960–2010)                     | GMM and 3SLS estimations                                                   | Non-linear (Disappearance of positive effects over time)                                                            | Correlation |
| Dabla-Noris and Srivisal (2013)             | 118 economies              | Panel data (1980–2010)                     | Dynamic panel analysis                                                     | Inverted U-shaped (High financial depth amplifies consumption and investment volatility)                            | Causality   |
| Law and Singh (2014)                        | 87 economies               | Panel data (1980–2010)                     | Dynamic panel threshold regressions (System GMM)                           | Inverted U-shaped (Identifies a finance-growth threshold at 88–100% of GDP)                                         | Correlation |
| Samargandi et al. (2015)                    | 52 middle-income economies | Panel data (1980–2008)                     | Panel Autoregressive Distributed Lag (ARDL) model and threshold regression | Inverted U-shaped link between finance and growth in the long run, but no significant short-run relationship.       | Correlation |
| Mamonov et al. (2018)                       | 63 economies               | Panel data (1980–2014)                     | Threshold regression and panel cointegration analysis                      | Non-linear (Various optimal financial sector depth identified; finance becomes destabilising beyond certain levels) | Correlation |
| Swamy and Dharani (2019)                    | 24 advanced economies      | Panel data (1983–2013)                     | Threshold and GMM regression                                               | Inverted U-shaped (Identifies a finance-growth threshold at 142% of GDP)                                            | Correlation |

*Source: authors' elaboration.*

## 4. Current landscape of the UK's financial sector

Building on the theoretical perspectives and empirical findings discussed in the literature review, this section examines the current state of the UK's FSS, focusing on its size, contribution to economic growth, and potential 'Financial Dutch Disease' characteristics.

### 4.1. Contribution to output and growth<sup>1</sup>

The UK ranks second among G7 economies in FSS GVA, behind only the United States (Figure 4.1). While Japan initially had a larger financial sector, the UK surpassed it around 2005. Germany, Canada, Italy, and France show steady but comparatively lower FSS growth, with France's FSS size declining in recent years. Relative to GDP, the UK and the US have the largest financial sectors at approximately 8% of total GDP, largely surpassing all other G7 peers (Figure 4.2). The trend underscores the UK's expanding FSS dominance relative to other major economies.

Furthermore, the FSS has faced two major shocks in recent years. The first was the GFC, which led to a decrease in the share of UK financial services GVA relative to GDP, which dropped by approximately 0.5 percentage points in 2007 (Figure 4.2). The second was Brexit, which caused the relocation of some 7,000 FSS jobs to mainland Europe to maintain European Single Market access (House of Lords, 2022), coupled with an 18% decrease in financial services exports to the EU (Hutton et al., 2024). Between 2016 and 2020, a period marked by Brexit negotiations, the share of FSS GVA relative to GDP decreased by almost one percentage point (Figure 4.2).

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<sup>1</sup> This section draws upon the ONS Blue Book 2024 dataset, presenting UK National Accounts data from 1987 to 2023 (ONS, 2024b).

**Figure 4.1.** Financial and insurance activities GVA (in real USD billions), UK vs G7  
(excluding US)



*Note: This figure excludes the US for scaling reasons, as its disproportionately large GVA levels make comparisons with other smaller G7 economies impossible.*

*Source: authors' elaboration using OECD data.*

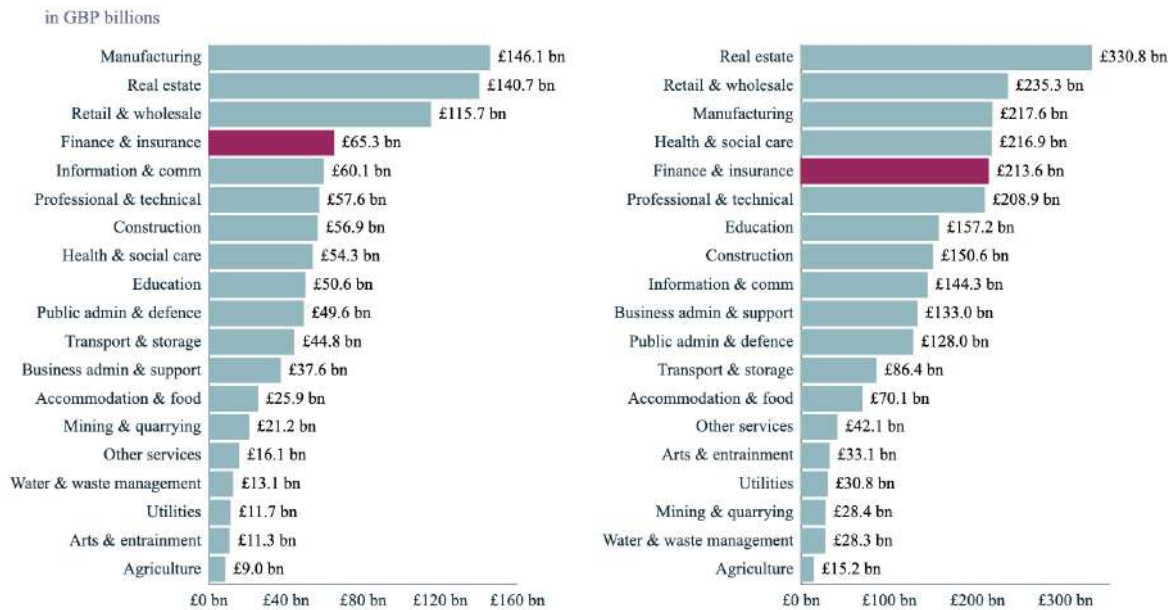
**Figure 4.2.** Financial and insurance activities GVA (share of GDP in %), UK vs G7



*Source: authors' elaboration using OECD data.*

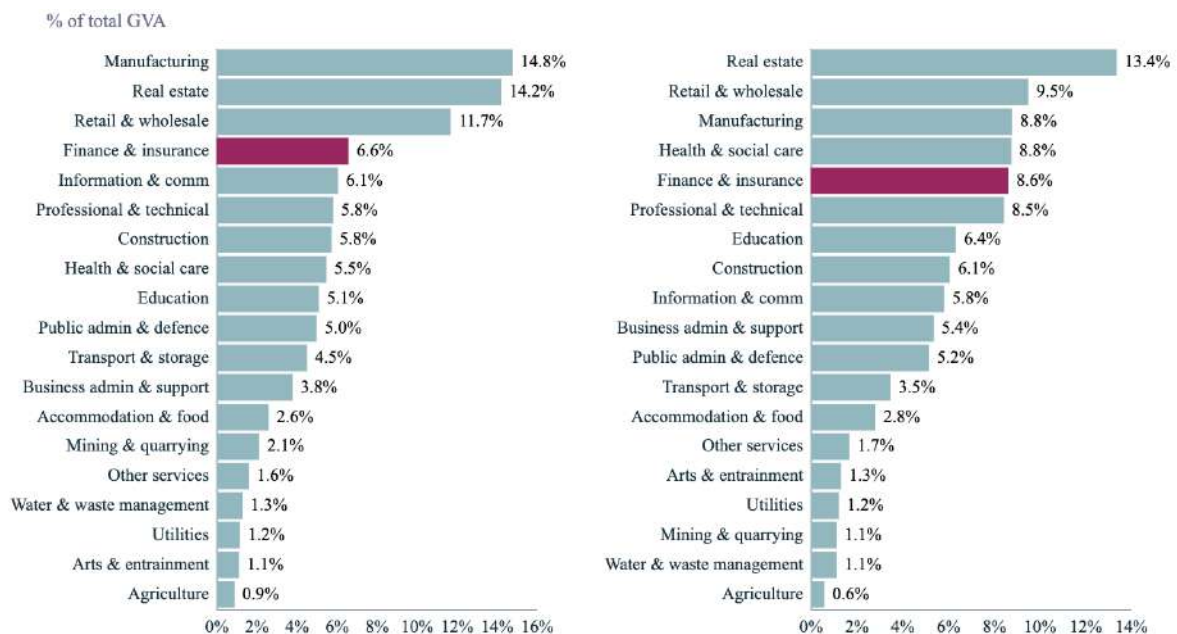
Focusing on the UK, the finance and insurance sector is the fifth largest industry in terms of GVA, at over £213 billion in 2024, up from £65 billion in 2000 (Figure 4.3). In 2024, financial services represented 8.6% of the total GVA in Britain, up from 6.6% in 2000, making it one of the highest contributing sectors to the UK economy today (Figure 4.4) (ONS, 2024b, 2025b).

**Figure 4.3.** UK GVA by industry (in real GBP billions at basic prices, 2019 money value), 2000 (left) vs 2024 (right)



Source: authors' elaboration using ONS GDP output approach low-level aggregates 2025.

**Figure 4.4.** UK GVA shares by industry (% of total GVA), 2000 (left) vs 2024 (right)



Source: authors' elaboration using ONS GDP output approach low-level aggregates 2025.

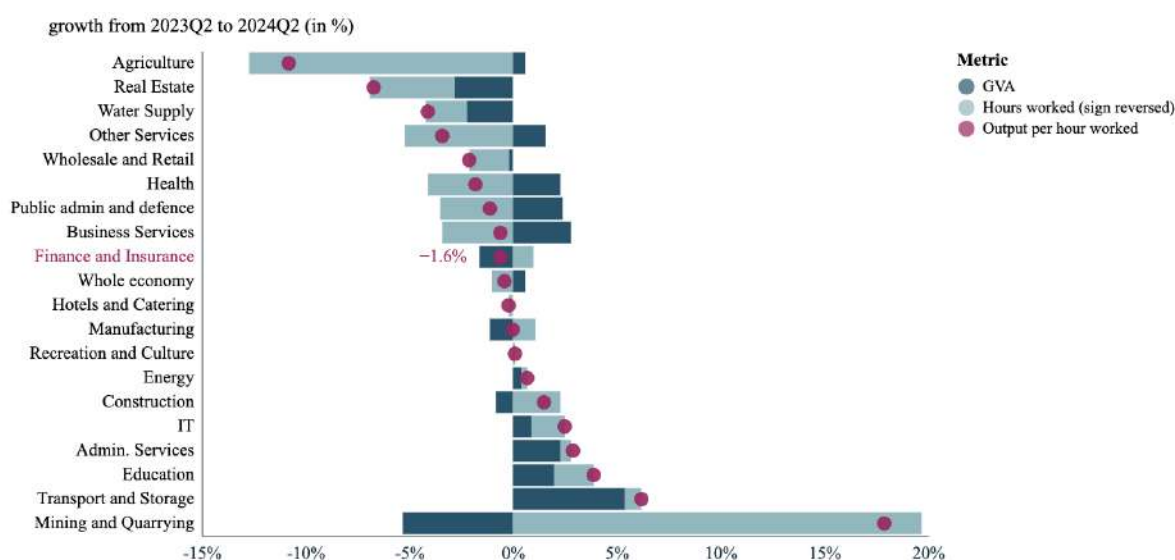
Overall, Britain's FSS size is undeniably large, the largest in developed economies along with the US. While impacted by subsequent crises (GFC, Brexit, Covid pandemic), the UK's financial sector is still a strong driver of economic output in the UK, at 8% of GDP.

## 4.2. Contribution to productivity

Building on the previous GVA analysis, this section examines the financial sector's contribution to overall productivity, measured in output per hour worked. Between Q2 2023 and Q2 2024, the UK's FSS recorded a negative -1.6% productivity growth, mainly driven by negative GVA growth (Figure 4.5). The intuition behind this is that a decrease in hours worked coupled with a decrease in GVA lowers output per hour worked. The impact would have been more pronounced if not for the decrease in total hours worked in the industry, which partially counteracts this effect.

Furthermore, the FSS recorded a negative -0.1% contribution to aggregate UK productivity in terms of output per hour worked (Figure 4.6). While many of the UK's largest industries contributed negatively or had no notable impact on productivity growth (manufacturing, business services, finance and insurance, health, wholesale, and retail), smaller industries appear to have been the main drivers of recent productivity growth (education, transport and storage, administrative services, ICT).

**Figure 4.5.** Industry decomposition of productivity growth (output per hour worked), UK 2023Q2 to 2024Q2 (in %)



Source: authors' elaboration using ONS productivity flash estimate data.

**Figure 4.6.** Industry contribution to aggregate productivity growth (output per hour worked) and relative industry size, UK 2023Q2 to 2024Q2 (in %)



Note: 'Hours worked (sign reversed)' means declines appear as positive values to highlight that falling hours, with stable or rising output, lead to higher productivity. For example, in mining and quarrying, a sharp drop in hours worked corresponded with strong productivity growth.

Source: authors' elaboration using ONS productivity flash estimate data.

Overall, it appears that a large part of the stagnating effects of FSS activity on total economic output has to do with negative FSS productivity growth. As the UK is faced with a large 'productivity gap' relative to G7 peers, lagging FSS productivity could be an explanation for the broader growth stagnation in Britain (Vaitilingham, 2024).

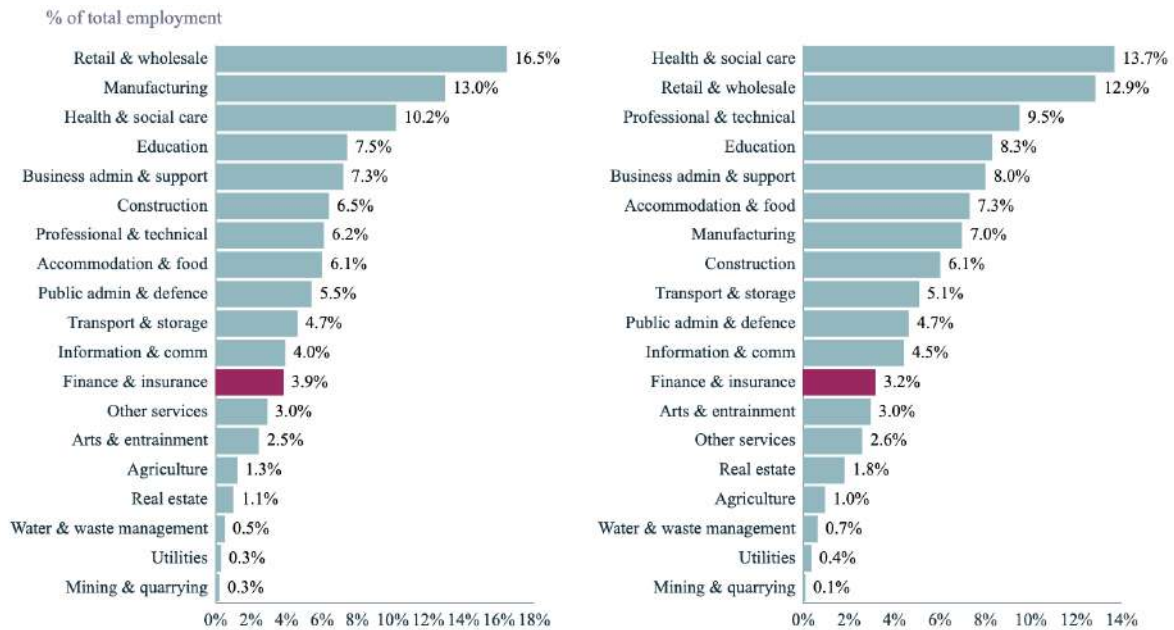
### 4.3. Contribution to employment

This section analyses Britain's FSS' contribution to total employment. By industry shares, the finance and insurance sector has a relatively low contribution to total employment, representing 3.2% of the 2024 workforce, down from 3.9% in 2000 (ONS, 2025c). The FSS has the eighth lowest industry contribution to employment in the UK, only slightly surpassing other service activities, arts and entertainment, agriculture, and real estate sectors (Figure 4.7). However, large FSS GVA combined with low FSS employment suggests that financial sector workers derive large output gains per worker, hinting at high FSS productivity.

A regional decomposition of financial sector employment reveals that Greater London concentrates close to 40% of total FSS employment in 2024, while the London region

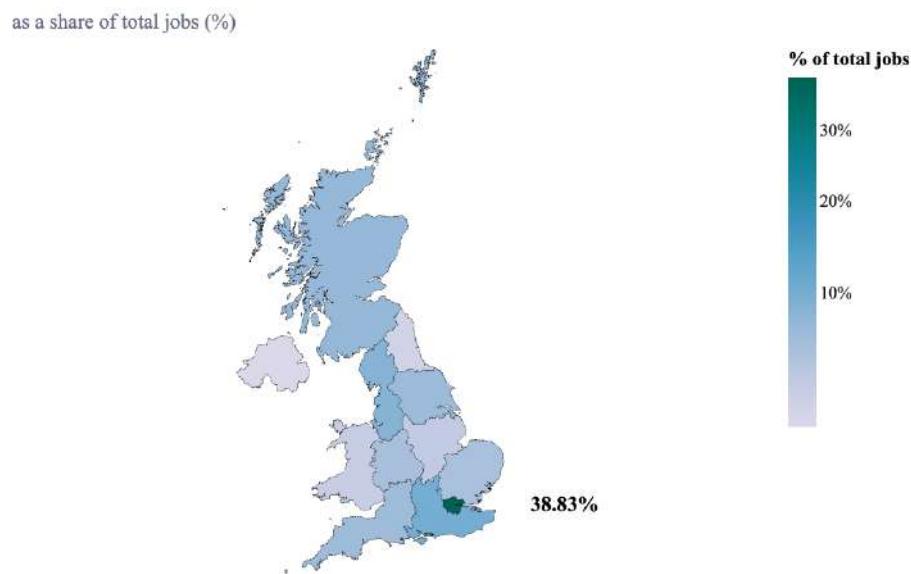
represents only 13% of the UK's population (Figure 4.8). In this regard, financial sector employment is disproportionately allocated towards Greater London, largely due to the prevalence of the City of London and Canary Wharf as global financial hubs. Low financial services sector employment outside London is consistent with the limited regional FSS GVA discussed earlier (Figures 2.1-2.2).

**Figure 4.7.** UK employment shares by industry (in %), 2000 (left) vs 2024 (right)



Source: authors' elaboration using ONS GDP output approach low-level aggregates 2025.

**Figure 4.8.** Finance and insurance employment share (in %) by UK ITL1 regions, 2024



Source: authors' elaboration using ONS Blue Book 2024 data.

Overall, evidence from this section does not support the idea that financial sector employment is crowding out jobs or economic growth in other industries, as proposed by Cecchetti & Kharroubi (2015). The sector's relatively small share of total employment suggests limited displacement of labour from other industries, particularly outside London, where FSS employment remains low.

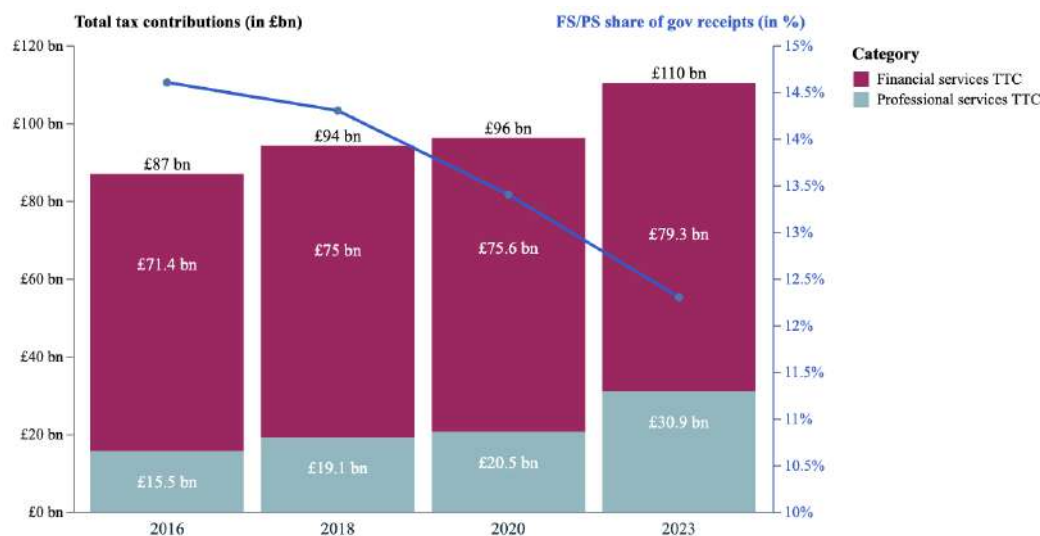
#### **4.4. Contribution to public revenue**

The FSS is unmistakably a significant contributor to UK total tax revenues. In 2023, UK FSS total tax contributions (TTC) amounted to £79.3 billion (Figure 4.9). Within the FSS, the sole banking sector contributed approximately £38 billion in tax receipts over the 2023-24 financial year, up from £21 billion in 2011-12 (Hutton et al., 2024). 2023-24 banking sector TTC are largely dominated by pay-as-you-earn taxes (at £25 billion) and corporation tax (at £10 billion).

However, while financial and related professional services represented close to 12% of total government tax receipts in 2023, their contribution to public revenue has declined by almost 3 percentage points since 2016 (Figure 4.9). The decline in FSS tax receipts as a share of government budget is to be interpreted with caution since it could simply result from increased tax revenues from other industries, altering the FSS' relative share.

Moreover, financial and insurance services are the largest contributing industry to UK corporation tax, at £18.8 billion in 2022-23 (PwC & TheCityUK, 2024) (Figure 4.10). Additionally, UK FSS represented 14% of all UK employment taxes in 2023 (at £55.8 billion), despite only representing 3.2% of total employment (PwC & TheCityUK, 2024).

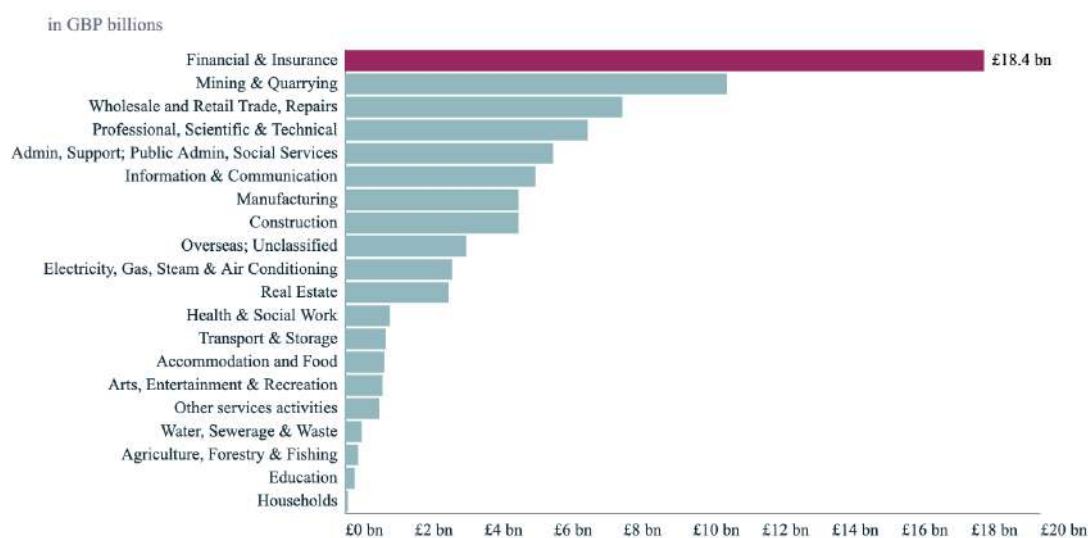
**Figure 4.9.** UK total tax contributions (TTC, in real GBP billions) and share of government receipts (in %) of financial and related professional services, 2016-23



*Note: Financial services are presented along with related professional services in figure following HM Revenue & Customs and HM Treasury accounting measures.*

*Source: authors' elaboration using The City UK and PwC data.*

**Figure 4.10.** UK total corporation tax receipts by industry, 2022-23 (in real GBP billions, 2019 money value)



*Source: authors' elaboration using TheCityUK and HMRC Corporation Tax Statistics data.*

Overall, UK financial and related professional services represented almost a sixth of total government receipts in 2023, making it one of the largest contributors to public revenue, both via corporate tax and income tax receipts. This estimate does not include VAT – likely

significant given the sector's high income and consumption levels – or capital gains taxes on investment returns, suggesting the sector's total fiscal contribution may be even greater.

#### 4.5. Financial sector financial flows

Given the financial sector's significant contributions to the UK economy through tax revenues, employment, and overall economic output, examining financial flows can help determine the extent to which this financial activity translates into domestic investment.

One way to examine this is by looking at the net acquisition of financial assets, which represents the total value of assets acquired by UK financial corporations minus those disposed of (Figure 4.11). These include cash, deposits, loans, bonds, equity, investment share funds, financial derivatives, and other receivable accounts. In 2023, UK financial corporations have acquired more assets abroad (at £226 billion) than in Britain (at -£83 billion) in nominal terms. This trend of shifting away from domestic investment toward international markets has been observable since 2020 and raises important questions about the attractiveness of UK financial assets and whether the FSS' outward investment strategy poses risks for long-term domestic economic growth.

Furthermore, the net acquisition of financial liabilities, defined as the total value of liabilities acquired by financial corporations minus those repaid or settled over the same period, offers additional insight into FSS financial flows (Figure 4.12). These include loans, debt securities, deposits, insurance technical reserves, financial derivatives and other payable accounts. Since the 2020 pandemic, the UK FSS has increasingly acquired liabilities from abroad and less so from the UK. In 2023, foreign liability acquisition reached £283 billion, nearly eight times higher than domestic liability acquisition (at £38 billion). As a result, the net acquisition of foreign liabilities has now reached its highest level since records began in 1987 (Figure 4.12).

**Figure 4.11.** UK financial corporations net acquisition of financial assets, 1998-2022 (in nominal GBP billions)



Source: authors' elaboration using ONS Blue Book 2024 data.

**Figure 4.12.** UK financial corporations net acquisition of financial liabilities, 1998-2022 (in nominal GBP billions)



Source: authors' elaboration using ONS Blue Book 2024 data.

Overall, the UK's FSS holds eight times more net financial obligations abroad than domestically. This reflects a broader trend of financial institutions shifting both assets and liabilities toward international markets. Combined, these figures challenge the conception that Britain's large FSS largely benefits the British economy. Empirically, it is undeniable that UK financial services largely contribute to economic output and economic growth, but these large financial flows are predominantly oriented towards the rest of the world's balance sheets.

#### 4.6. Contribution to productive investment

Delving deeper into FSS domestic financial flows, financial Gross Fixed Capital Formation (GFCF) measures the value of investments made by financial corporations in fixed assets, providing insights into the contribution of the FSS towards productive investment in the economy. Over the 1997-2023 period, financial GFCF has close to doubled, after reaching a global minimum of £72 billion in 2009 in the aftermath of the GFC (Figure 4.13). It reached its global maximum at over £174 billion just before the 2020 pandemic, sharply declining until 2021. Since then, GFCF has been slowly recovering, attaining just over £139 billion in 2023.

Relative to its G7 peers, UK financial GFCF represents just over 0.6% of GDP (Figure 4.14). This is in line with Italy, Japan, Canada and Germany, while remaining largely below the US and France's GFCF to GDP levels, at 1.2% and 0.8% respectively. In this regard, while the UK's FSS is the largest in its G7 peer group, its contribution to total investment in the economy is strikingly lower than its large size would suggest. For instance, at a similar FSS GVA-to-GDP ratio, the US financial sector contributes over twice as much to investment.

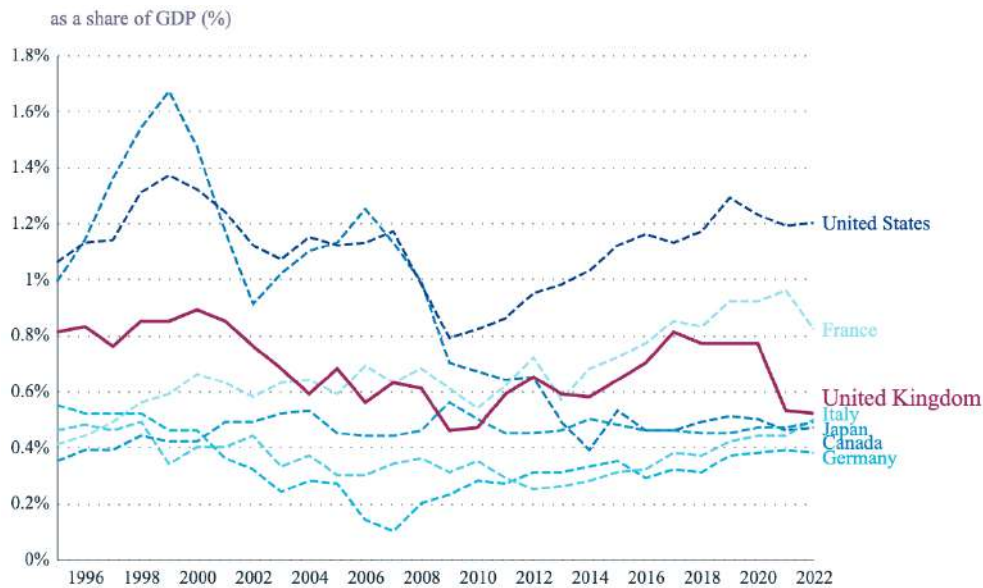
**Figure 4.13.** UK financial corporations GFCF (in current GBP billions, purchasers' prices)



*Note: This figure uses current prices as chained prices are not available by sectoral differentiation in the ONS Blue Book 2024.*

*Source: authors' elaboration using ONS Blue Book 2024 data.*

**Figure 4.14.** Financial corporations GFCF (share of GDP in %), UK vs G7



Source: authors' elaboration using OECD data.

A breakdown of FSS investment by sector reveals a long-term increase in intangible assets investment, with intellectual property products receiving the largest share of FSS GFCF in 2023 (at £59 billion), followed by ICT (at £42 billion). However, investment in tangible assets has largely fluctuated over time, marked by high volatility in FSS GFCF towards transport equipment (at £19 billion) and buildings and infrastructure (at £17 billion) (Figure 4.15).

**Figure 4.15.** UK financial corporations GFCF by sector (in current GBP billions at purchasers' prices)



Note: This figure uses current prices as chained prices are unavailable by sector in the ONS Blue Book 2024.

Source: authors' elaboration using ONS Blue Book 2024 data.

Overall, it appears UK financial sector investment is lower than its high GVA would suggest, relative to G7 peers. From a national perspective, the recent sharp decline in ICT and buildings and infrastructure investment could be an explanation for the UK's 'chronic levels of underinvestment', widely regarded as one of the main reasons for decades of stagnating growth (Chadha & Venables, 2024). In this context, incentivising financial sector financial flows towards productive investments should be prioritised to foster sustained future growth.

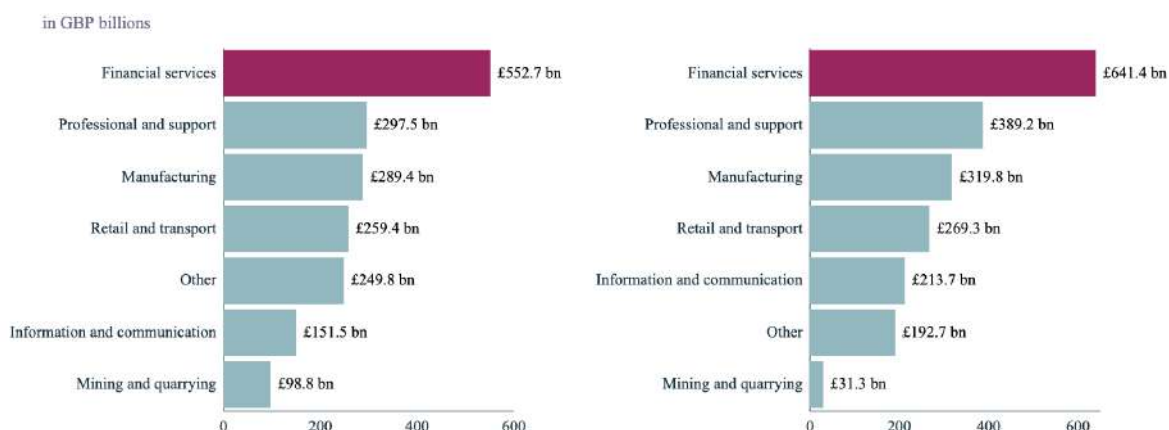
#### **4.7. Foreign direct investment and international trade**

A key characteristic of 'Financial Dutch Disease' is the flow of large foreign direct investment (FDI) into a single sector, potentially leading to currency appreciation and economic structural imbalances. In the UK, the financial sector plays a central role in attracting foreign capital, reinforcing its economic dominance.

In 2023, financial services accounted for the largest share of FDI inward flows into the UK, receiving nearly 30% of total foreign investment (at £641 billion), followed by professional and support services at 25% (Figure 4.16) (ONS, 2025a). Moreover, earnings from financial services FDI increased twofold from 2019 to 2022 (at £16 billion versus £32.6 billion respectively) (Hutton et al., 2024). This suggests that foreign investors increasingly favour finance over other sectors, further concentrating capital inflows to the UK in financial services.

However, regional disparities are also prevalent in FDI flows, as London captures 45% of total foreign investment while only representing 22% of the British GDP (ONS, 2024a). As such, financial FDI inflows are highly unequally distributed, further suggesting that FSS expansion may have reinforced sectoral imbalances and deepened regional economic divides.

**Figure 4.16.** Inward foreign direct investment (FDI) positions by industry, 2020 (left) vs 2023 (right) (in current GBP billions)



*Note: In line with international guidance, UK FDI statistics are presented in current prices on an immediate partner country basis.*

*Source: authors' elaboration using ONS FDI 2025 data.*

Beyond FDI, financial services are the second-largest export industry in Britain. In 2024, UK FSS exports amounted to £91.8 billion out of a total of £838 billion in real terms, with a net trade surplus of £73.2 billion. This represents approximately 12% of total British exports (Hutton et al., 2024).

However, unlike classic Dutch Disease patterns, there is limited evidence of financial sector expansion driving currency appreciation. Sterling has depreciated since 2007, from 2 USD per GBP before the GFC to 1.5 USD in the early 2010s and 1.3 USD in 2016 after Brexit negotiations (Investing UK, 2025).

Overall, while Sterling's decline suggests the UK may not be concerned by the traditional Financial Dutch Disease framework, the dominance of the FSS may still expose the British economy to other 'finance curse' effects. There may be alternative channels through which a potentially oversized financial sector can impose economic costs on the broader economy, raising the question of whether there is an optimal threshold beyond which financial sector growth ceases to be beneficial. Accordingly, examining these dynamics is essential to determining whether the financial sector has exceeded a sustainable size and if further expansion risks harming broader economic performance. The following sections in this study address this important question.

## 5. Research design and methodology

### 5.1. Conceptual framework

The UK's government's prime desire to kickstart economic growth in the UK has seen growth re-enshrined as the FCA's secondary objective (Rathi, 2024b; Reeves, 2024). In times when the UK is predicted to be the worst-performing G7 economy in 2025, understanding the role of financial services in driving sustained and sustainable growth in the country is crucial (Inman, 2024).

However, as mentioned in their research note on 'The Growth Gap', the FCA has highlighted the need to better understand the contribution of the FSS to enhance competitiveness and growth in the UK (Morrison et al., 2024). As discussed in Section 2, this study aims to answer the following research question: *'Is the UK's financial services sector disproportionately large?'*

This study's research design closely builds on that of Cecchetti & Kharroubi (2012). The paper challenges the common idea that financial development is good for growth, arguing that 'like a person who eats too much, a bloated financial system [may] become a drag on the rest of the economy' (Cecchetti & Kharroubi, 2012, p. 1). The underlying assumption behind this reasoning is that the financial sector competes for resources with the broader economy, i.e. physical capital and, in particular, highly skilled human capital. As such, 'people who in another age dreamt of curing cancer or flying to Mars, today dream of becoming hedge fund managers', potentially holding down aggregate productivity and economic growth (Cecchetti & Kharroubi, 2012, p. 2).

### 5.2. Hypothesis

This study challenges the hypothesis that financial development has a positive effect on economic growth up to a certain point, after which returns are negative, in line with the empirical results found in the aforementioned papers. That is, the estimator for  $b_1$  is positive and the estimator for  $b_2$  is negative (estimators are defined in Section 5.3). Because of the large heterogeneity in treatment effects estimated in the literature, this study questions the robustness of these results.

### 5.3. Methodological approach

This research is motivated by the literature showing an inverted U-shape relationship between finance and growth (Arcand et al., 2012; Cecchetti & Kharroubi, 2012). Guided by these studies, this study employs panel regression analysis, regressing GDP per capita growth rates on financial development, squared financial development, output per capita and several control variables.

First, this study replicates the methodological design of Cecchetti & Karroubi (2012), allowing for the same analysis to be conducted over a longer time frame. This extension tests whether the original findings remain consistent over an extended period. To ensure proper comparability, estimates consider the same group of countries as the authors (for more details on country samples see Annex 11.2).

Second, this research explores the robustness of results by exploring the limitations of the paper's methodological groundings and functional forms. In particular, this research studies outliers and includes country-specific and time-specific fixed effects, a serious omission from the original study. As such, this paper proposes a revised methodological design to model the effect of private credit to GDP and financial employment growth on economic growth.

This study is an important contribution to the existing literature, as it updates scholarly findings to the most recently available data, verifies and challenges the robustness of existing identification strategies, and derives conclusions from these core findings.

The main regression is estimated using ordinary least squares (OLS) and follows this form:

$$\Delta y_{i,t,t+5} = \alpha + \beta_i + \gamma_t + b_1(fd_{i,t,t+5}) + b_2(fd_{i,t,t+5})^2 + b_3X_{i,t,t+5} + \delta y_{i,t} + \varepsilon_{i,t} \quad \text{Eq. 1}$$

Where:

- $y_{i,t}$  is the log GDP per capita in country  $i$  and year  $t$ ;
- $\Delta y_{i,t,t+5}$  is the average GDP per capita growth in country  $i$  from year  $t$  to  $t + 5$ ;
- $fd_{i,t,t+5}$  is the average level of financial development in country  $i$  from  $t$  to  $t + 5$ ;

- $X_{i,t,t+5}$  is the set of control variables averaged from  $t$  to  $t + 5$ ;
- $\alpha$  is the intercept (constant);
- $\beta_i$  is the vector of country dummies;
- $\gamma_t$  is the vector of year dummies, only included in Section 6.2;
- $\varepsilon_{i,t}$  is the error term, controlling for heteroskedasticity in Section 6.1, and clustering the errors in Section 6.2;
- $b_1$  and  $b_2$  are the estimator coefficients for the relationship between financial development and growth.
- $b_3$  is the estimator for control variables' treatment effects.

To determine the inflexion point of the parabola, one must calculate where the financial sector is at its optimal size. To do so, the regression is differentiated with respect to  $fd_{i,t,t+5}$ , as given below:

$$\frac{\partial \Delta y_{i,t,t+5}}{\partial fd_{i,t,t+5}} \Leftrightarrow b_1 + 2b_2(fd_{i,t,t+5}) = 0 \Leftrightarrow fd_{max} = -\frac{b_1}{2b_2} \quad \text{Eq. 2}$$

Where  $fd_{max}$  is the level of financial development that maximises GDP per capita growth. Then,  $fd_{max}$  can be plugged into the initial regression to determine  $\Delta y_{max}$ , i.e. the maximum GDP per capita growth rate obtained when the financial sector is at its 'optimal size'. Thus, the cost to growth from having 'too much finance' can be calculated as the difference between maximum growth  $\Delta y_{max}$  and the observed average growth rate for the data series. Nevertheless, it is important to note that this equation is only valid if one expects to find positive returns to financial development and then negative results. That is, the estimator for  $b_1$  is positive and the estimator for  $b_2$  is negative. However, true estimates could be significantly different, as observed in Section 6.

## 5.4. Metrics for financial development

This research builds on the following metrics as proxies for financial services sector size and development.

### 5.4.1. Private credit to GDP

Private credit (by deposit money banks and other financial institutions) to GDP measures the total amount of private credit provided to the private sector by financial institutions, relative to GDP. The higher the ratio, the larger the financial sector. Private credit to GDP is an important metric because it measures the ability of businesses and households to access financing for investment and consumption, a relevant component to GDP growth.

#### **5.4.2. Private credit by banks to GDP**

An alternative measure of financial sector size is private credit by deposit money banks rather than total credit. The main motivation behind including this metric is that ‘total credit’ is ambiguous depending on country-specific contexts. In particular, while some countries have bank-based financial systems (where commercial banks play the main role in providing credit), others have market-based financial systems (where stock and bond markets play a major role in providing credit to the private sector). As such, the relationship between ‘total credit’ and economic growth may look different simply due to the context-specific composition of the credit system. For this reason, running analyses solely on private credit by banks enables minimising these comparability issues.

#### **5.4.3. Financial sector share in total employment**

Measuring financial development using private credit focuses on outputs. An alternative measurement is to focus on inputs, i.e. the ratio of financial sector employment to total employment in the economy. This measures the sector’s use of labour resources, useful in determining potential ‘Dutch Disease’ effects of potentially crowding out other sectors. As such, a higher employment share reflects a greater weight of the financial sector in the use of production inputs.

### **5.5. Data sources**

To examine the relationship between financial development and economic growth, this paper combines three different datasets (for more information on data collection, see Annex 11.1).

First, information on the size of the financial sector is obtained from the Global Financial Development Database (World Bank, 2022). The two main variables extracted are private

credit ratio (defined as the ratio of private credit by deposit money and other financial institutions to GDP) and private credit by banks ratio (defined as the ratio of private credit by deposit money institutions to GDP). The share of financial and insurance activities in total employment is constructed using data from the Structural Analysis Industry Database (OECD, 2025b) and therefore is only available for OECD countries.

Second, information on economic performance and other country-level macroeconomic data used to control for country-specific and time-specific fixed effects is obtained from the World Development Indicators (World Bank Data, 2023). These variables include GDP per capita (constant 2015 USD), GDP per capita annual growth, the ratio of trade to GDP, inflation, government expenditure as a share of GDP, and the working age population which is used to calculate working age population growth. Data on the regulatory quality is obtained from the Worldwide Governance Indicators (World Bank, 2025b).

The merged dataset covers the period from 1960 to 2024, though data availability varies across years and countries. For instance, the private credit ratio is available only until 2017, while private credit by banks extends until 2021. Descriptive statistics of the working dataset are given below (Table 5.1). Five-year averages non-overlapping are then generated for all variables except for the logarithm of GDP per capita.

**Table 5.1. Working dataset descriptive statistics**

| Variable name                 | Observations | Mean     | Std. Dev. | Min       | Max      |
|-------------------------------|--------------|----------|-----------|-----------|----------|
| Log of GDP Per Capita         | 11,689       | 8.35617  | 0.01383   | 4.80957   | 12.322   |
| GDP Per Capita Growth         | 10,858       | 1.89314  | 0.06148   | -64.4236  | 140.4906 |
| Finance Employment Share      | 844          | 0.02748  | 0.00974   | 0.00816   | 0.06190  |
| Private Credit Ratio          | 8,142        | 36.25872 | 40.48225  | 0.00677   | 385.8848 |
| Private Credit by Banks Ratio | 8,511        | 37.41967 | 34.92285  | 0.01037   | 304.5745 |
| Population Working Age Growth | 14,127       | 1.98554  | 2.04249   | -30.76747 | 30.3498  |
| Openness to Trade             | 7,868        | 77.08862 | 51.89833  | 0         | 863.1951 |
| Government Consumption        | 8,276        | 16.51936 | 8.18781   | 0         | 147.735  |
| CPI Inflation                 | 8,483        | 23.70363 | 328.0775  | -17.64042 | 23773.13 |
| Regulatory Quality            | 4,924        | 49.49196 | 29.0088   | 0         | 100      |

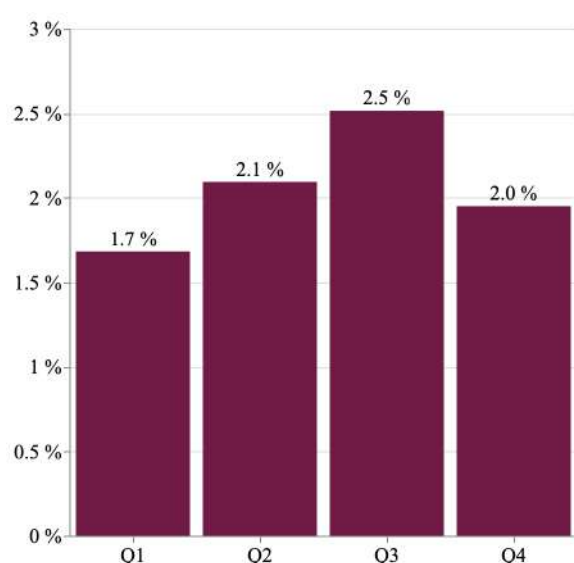
*Source: authors' elaboration.*

A simple histogram is constructed by computing average GDP per capita growth by bank's private credit ratio quartiles. The resulting histogram, based on 1,863 data points, reveals that

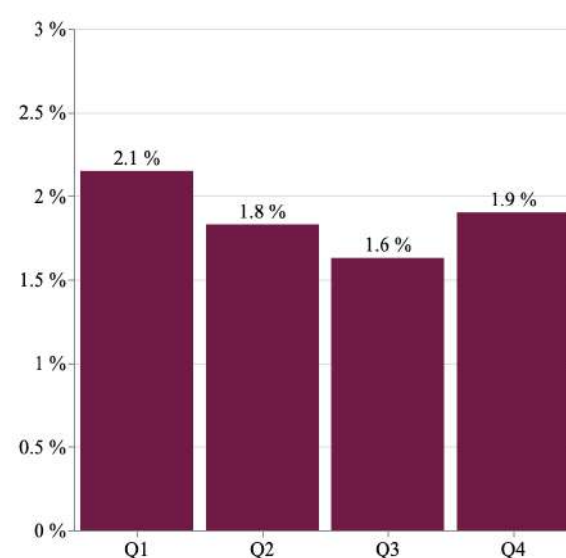
GDP per capita growth increases from the first to the third quartile and declines in the fourth quartile (Figure 5.1). In other words, countries with the highest levels of private credit by banks to GDP experience on average lower growth than those in the intermediate quartiles.

This pattern doesn't hold when plotting average GDP per capita growth by financial employment quartiles (Figure 5.2.), which due to data limitations, include only OECD countries.

**Figure 5.1.** Average GDP per capita growth by private credit by banks to GDP quartiles



**Figure 5.2.** Average GDP per capita growth by financial employment quartiles



*Source: authors' elaboration.*

## 5.6. Methodological limitations

This section examines the limitations in Cecchetti & Karroubi's (2012) methodology, which consequently also applies to our first set of regression estimates that replicate their results. They are also true for the histograms presented above.

Although the histograms illustrate a correlation between private credit by banks to GDP and GDP per capita growth, they cannot confirm whether exceeding a certain threshold of financial credit negatively affects economic growth. The same applies to this study's regression analysis design described in Section 5.3. Several confounding factors could be influencing this pattern, including omitted variables that drive both credit expansion and GDP growth, making it unclear

whether the observed relationship is due to the financial sector itself or to broader economic conditions.

Moreover, the analysis does not account for potential endogeneity issues. High private credit to GDP ratios may be the result of slower economic growth rather than its cause, as countries with stagnating economies may rely more on credit to sustain investment and consumption. Additionally, different institutional and macroeconomic environments across countries can affect how financial credit interacts with growth, making it difficult to generalise findings. Without controlling for these factors through more rigorous econometric techniques, this study alone cannot support the claim that, beyond a certain threshold, financial credit directly hinders GDP per capita growth.

Furthermore, this design does not account for unobserved, time-invariant characteristics of individual countries or broader global trends affecting all countries simultaneously. Structural differences, such as financial regulations, political stability, and long-term growth potential, could be influencing both credit expansion and economic performance. Similarly, global economic cycles, crises, or major policy shifts could be driving changes in both variables. Without these controls, the observed relationship might be driven by country-specific or time-specific factors rather than a fundamental economic mechanism linking financial credit and growth.

The econometric approach described in Section 6.2 addresses this issue by incorporating country and year-fixed effects, but it does not resolve the other aforementioned concerns. The next section presents the results of this study.

## 6. Study results

### 6.1. Replication of results

This section replicates the analysis conducted by Cecchetti & Kharroubi (2012) in estimating a model that closely follows their specification, including the use of the same sample of countries (for more country sample details see Annex 11.2). The time frame begins in 1980, consistent with the original study, and extends to the most recent year for which data are available, varying by variable. Following Cecchetti & Kharroubi's (2012) approach, standard errors are robust to heteroskedasticity but do not account for autocorrelation, a limitation that could be artificially inflating the significance of the coefficients. This issue is addressed in the next section.

Table 6.1. presents the results using the private credit-to-GDP ratio as the independent variable. The first column shows the simplest specification, with no controls. Subsequent columns sequentially introduce additional controls, such as working-age population growth, openness to trade, government consumption, and inflation. Column (6) replicates the specification of column (5) while adding country-fixed effects, and column (7) further incorporates year-fixed effects.

In columns (1) through (5), the relationship between private credit and economic growth appears to follow a parabolic pattern, evidenced by a significantly positive coefficient on the linear term  $b_1$  and a significantly negative coefficient on the quadratic term  $b_2$ . However, columns (6) and (7) show that once fixed effects are included, the coefficients on private credit lose significance, suggesting that part of its impact may be driven by unobserved country-specific factors.

Amongst controls, log GDP per capita consistently shows a negative and significant effect, supporting the conditional convergence hypothesis. Trade openness and population growth contribute positively to growth in some specifications, while government consumption is strongly associated with lower growth. Inflation does not have a statistically significant effect. The R-squared increases from 0.24 to 0.64 as more controls and fixed effects are introduced, highlighting the importance of accounting for unobserved heterogeneity in explaining GDP per capita growth.

**Table 6.1.** Regression results: GDP per capita growth and private credit (5-year averages)  
using Cecchetti & Kharroubi's (2012) sample

|                  | (1)        | (2)        | (3)        | (4)        | (5)        | (6)        | (7)         |
|------------------|------------|------------|------------|------------|------------|------------|-------------|
| Priv. Credit     | .05511***  | .05338***  | .04354***  | .04672***  | .04081***  | .01862     | .00579      |
| (5y)             | (.01268)   | (.01338)   | (.01375)   | (.01457)   | (.01416)   | (.01936)   | (.01723)    |
| Squared Priv.    | -.00028*** | -.00027*** | -.00022*** | -.00024*** | -.00022*** | -.00013    | -.00011     |
| Credit (5y)      | (.00006)   | (.00006)   | (.00007)   | (.00007)   | (.00007)   | (.00009)   | (.00008)    |
| Log Real         | -.71433*** | -.8128***  | -.8832***  | -.65865*** | -.61386*** | -1.15596** | -2.23072*** |
| GDP pc           | (.12248)   | (.13573)   | (.13344)   | (.15719)   | (.15801)   | (.52331)   | (.63855)    |
| Pop. Growth      |            | -.21046    | -.16157    | -.36912*   | -.37156*   | -.51706*   | -.24488     |
| (5y)             |            | (.19088)   | (.19896)   | (.19517)   | (.19481)   | (.27332)   | (.27053)    |
| Trade Openness   |            |            | .00729**   | .00642**   | .0055**    | .0095      | -.00307     |
| (5y)             |            |            | (.00286)   | (.0028)    | (.0027)    | (.0073)    | (.00853)    |
| Gov. Consumption |            |            |            | -.12221*** | -.13048*** | -.29851*** | -.29201***  |
| (5y)             |            |            |            | (.03349)   | (.03213)   | (.08801)   | (.07903)    |
| CPI Inflation    |            |            |            |            | -.00324    | -.00347    | -.00214     |
| (5y)             |            |            |            |            | (.00232)   | (.00273)   | (.00224)    |
| Observations     | 375        | 375        | 364        | 362        | 354        | 354        | 354         |
| R-squared        | .14334     | .14885     | .17455     | .2049      | .2306      | .49289     | .57215      |
| Country FE       | NO         | NO         | NO         | NO         | NO         | YES        | YES         |
| Year FE          | NO         | NO         | NO         | NO         | NO         | NO         | YES         |

*Standard errors are in parentheses*

\*\*\*  $p < .01$ , \*\*  $p < .05$ , \*  $p < .1$

*Source: author's elaboration.*

Table 6.2. reports similar results using the ratio of private credit provided by deposit money banks to GDP as the independent variable. The modelling approach is identical, beginning with a specification without controls and progressively adding controls. The findings are consistent with those in Table 6.1. A significant parabolic relationship is observed in the initial specifications, which disappears when country and year-fixed effects are incorporated. The key difference in this case is that the linear term  $b_1$  becomes negative, although statistically insignificant.

**Table 6.2.** Regression results: GDP per capita growth and private credit by banks (5-year averages) using Cecchetti & Kharroubi (2012) sample

|                                       | (1)                    | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    | (7)                    |
|---------------------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| Priv. credit by banks<br>(5y)         | .03946***<br>(.01172)  | .03317***<br>(.01201)  | .02765**<br>(.01153)   | .02873**<br>(.0128)    | .02578*<br>(.01338)    | -.00519<br>(.01402)    | -.01196<br>(.01392)    |
| Squared priv. credit<br>by banks (5y) | -.0002***<br>(.00006)  | -.00017***<br>(.00006) | -.00014**<br>(.00006)  | -.00015**<br>(.00007)  | -.00014**<br>(.00007)  | -.00003<br>(.00007)    | -.00001<br>(.00007)    |
| Log real GDP pc                       | -.66137***<br>(.11639) | -.79024***<br>(.13015) | -.95217***<br>(.13085) | -.75967***<br>(.15232) | -.73109***<br>(.15336) | -1.23423**<br>(.5005)  | -1.80905***<br>(.5492) |
| Pop. growth<br>(5y)                   |                        | -.32921**<br>(.13086)  | -.36764**<br>(.14975)  | -.54887***<br>(.15028) | -.56281***<br>(.14972) | -.66828***<br>(.21827) | -.46772**<br>(.21722)  |
| Trade openness<br>(5y)                |                        |                        | .00938***<br>(.00262)  | .00886***<br>(.00254)  | .00813***<br>(.00246)  | .01194*<br>(.00698)    | .00666<br>(.00762)     |
| Gov. consumption<br>(5y)              |                        |                        |                        | -.10939***<br>(.03301) | -.1221***<br>(.03143)  | -.26862***<br>(.07785) | -.26148***<br>(.07168) |
| CPI inflation<br>(5y)                 |                        |                        |                        |                        | -.00141***<br>(.00031) | -.00087<br>(.00067)    | -.00023<br>(.00064)    |
| Observations                          | 411                    | 411                    | 356                    | 354                    | 346                    | 346                    | 346                    |
| R-squared                             | .11928                 | .13459                 | .20097                 | .22846                 | .24336                 | .56299                 | .61648                 |
| Country FE                            | NO                     | NO                     | NO                     | NO                     | NO                     | YES                    | YES                    |
| Year FE                               | NO                     | NO                     | NO                     | NO                     | NO                     | NO                     | YES                    |

*Standard errors are in parentheses*

\*\*\*  $p < .01$ , \*\*  $p < .05$ , \*  $p < .1$

*Source: author's elaboration.*

Table 6.3. presents the results using the share of financial employment as the proxy for financial sector size. Cecchetti & Kharroubi (2012) reduce the sample of countries when using this variable and this table does the same, leading to a sharp drop in observations. Similar to the previous tables, the inverted U-shape is statistically significant only in the first two columns but disappears in the subsequent ones. Notably, in column (7),  $b_1$  is negative and statically significant, pointing to a negative correlation between finance and growth.

**Table 6.3.** Regression results: GDP per capita growth and share of employment in the financial sector (5-year averages) using Cecchetti & Kharroubi's (2012) sample

|                    | (1)         | (2)         | (3)         | (4)         | (5)         | (6)         | (7)         |
|--------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Finance            | 159.3129*** | 143.1075**  | 127.3667*   | 73.8125     | 55.8620     | -279.9772   | -306.0744*  |
| employment (5y)    | (59.1848)   | (59.6484)   | (75.0123)   | (71.3979)   | (74.4384)   | (180.5062)  | (154.2598)  |
| Squared finance    | -1769.859** | -1654.654** | -1421.0254  | -754.54415  | -645.39191  | 1639.9551   | 1825.9658   |
| employment (5y)    | (731.68016) | (736.7743)  | (883.41193) | (832.33315) | (890.98298) | (2394.3878) | (1929.4232) |
| Log real GDP pc    | -2.2651***  | -2.17283*** | -2.25736*** | -.67663     | -1.68881*** | -4.76576*** | -3.67554*** |
|                    | (.45304)    | (.44429)    | (.49464)    | (.54086)    | (.48269)    | (.80224)    | (.99734)    |
| Pop. growth (5y)   |             | .21862      | .27285      | -.05151     | .31975      | -.38759     | -.03992     |
|                    |             | (.18114)    | (.20496)    | (.17744)    | (.23158)    | (.28987)    | (.27775)    |
| Trade openness     |             |             | .00333      | .00608*     | .01121***   | .0083       | .00895      |
| (5y)               |             |             | (.00302)    | (.00306)    | (.00353)    | (.01223)    | (.01031)    |
| Gov. consumption   |             |             |             | -.10428**   | -.20454***  | -.55102***  | -.43802***  |
| (5y)               |             |             |             | (.04901)    | (.04882)    | (.10699)    | (.09187)    |
| CPI inflation (5y) |             |             |             |             | -.07289     | -.13502***  | -.03135     |
|                    |             |             |             |             | (.05367)    | (.04507)    | (.04818)    |
| Observations       | 117         | 117         | 103         | 60          | 103         | 103         | 103         |
| R-squared          | .30318      | .31018      | .30318      | .2384       | .41703      | .71185      | .83269      |
| Country FE         | NO          | NO          | NO          | NO          | NO          | YES         | YES         |
| Year FE            | NO          | NO          | NO          | NO          | NO          | NO          | YES         |

*Standard errors are in parentheses*

\*\*\*  $p < .01$ , \*\*  $p < .05$ , \*  $p < .1$

*Source: author's elaboration.*

The findings summarised in the three previous tables suggest that the relationship between financial sector size and economic growth lacks robustness when alternative measures or control variables are used. The variation in results across different specifications indicates that the observed effects are highly sensitive to model specification and variable selection. These inconsistencies raise concerns about the reliability of the estimated relationship and limit the strength of any definitive conclusions.

## 6.2. Own analysis from methodological refinements

This section builds on the main specification model while deviating from Cecchetti & Kharroubi's (2012) approach in several key aspects.

First, unlike Cecchetti & Kharroubi (2012), the sample is not restricted to a specific set of countries. Instead, all countries with available data are included, as there is no clear justification for excluding some. Additionally, the analysis begins in 1960, the earliest year with data availability, and extends to the most recent period.

Second, this section explores only two measures of financial development: private credit by banks relative to GDP and financial employment share. Private credit by banks is preferred over total private credit, as it provides more recent data coverage, whereas total private credit-to-GDP is only available until 2018.

Regarding controls, all estimations incorporate country and year-fixed effects. However, openness to trade and government consumption are excluded as control variables. Indeed, financial development (measured by private credit) can facilitate trade by improving access to financing for exporters and importers. If private credit affects trade, and trade affects GDP growth, then the variable is a bad control because it lies on the causal path of treatment effects. Government consumption as a share of GDP can also be a problematic control. Financial development influences government spending by improving access to credit and increasing revenues, while also affecting the composition of expenditures (Chen et al., 2019; Kapaya, 2023).

Instead, a new control for regulatory quality is introduced, though data is only available after 1996. This variable aims to capture the idea that the impact of the financial sector on economic growth may depend not only on its size but also on the effectiveness of its regulation, as explored in greater depth in previous sections.

Additionally, all estimations in this section cluster standard errors, addressing a methodological limitation in the original study, where authors did not account for within-country correlation in errors. Clustering standard errors at the country level is essential in this analysis to account for within-country correlation over time, ensuring more reliable inference. Economic growth and financial sector variables tend to exhibit persistence, leading to serial correlation that violates the assumption of independent errors. Additionally, heteroskedasticity across countries and shared institutional or policy effects can bias standard errors if not addressed. Without clustering, standard errors may be underestimated, inflating t-statistics and overstating significance.

Table 6.4. presents the results of these estimations, using the ratio of private credit by banks to GDP as the dependant variable. In the first three specifications, both  $b_1$  and  $b_2$  are statistically insignificant. In column (4), where the sample size is significantly reduced by the incorporation of the regulatory quality control,  $b_1$  becomes negative and weakly significant while  $b_2$  is still not significantly different from zero.

**Table 6.4.** Regression results: GDP per capita growth and private credit by banks (5-year averages) using the full sample (countries and years)

|                                    | (1)                    | (2)                     | (3)                     | (4)                   |
|------------------------------------|------------------------|-------------------------|-------------------------|-----------------------|
| Priv. Credit by Banks (5y)         | .00748<br>(.01331)     | .00598<br>(.01353)      | -.01673<br>(.01264)     | -.05052**<br>(.02499) |
| Squared Priv. Credit by Banks (5y) | -.00009<br>(.00006)    | -.00009<br>(.00006)     | .00000<br>(.00005)      | .00011<br>(.00008)    |
| Log Real GDP pc (US\$)             | -2.54174***<br>(.5338) | -2.53773***<br>(.53011) | -2.39906***<br>(.68478) | -3.33033<br>(2.09737) |
| Pop. Growth (5y)                   |                        | -.17675<br>(.17637)     | -.31175<br>(.19052)     | -.08138<br>(.20066)   |
| CPI Inflation (5y)                 |                        |                         | -.00403***<br>(.00103)  | -.01487<br>(.00968)   |
| Regulatory quality                 |                        |                         |                         | -.00313<br>(.01902)   |
| Observations                       | 1717                   | 1717                    | 1460                    | 786                   |
| R-squared                          | .30057                 | .30242                  | .36165                  | .45099                |
| Country FE                         | YES                    | YES                     | YES                     | YES                   |
| Year FE                            | YES                    | YES                     | YES                     | YES                   |

*Standard errors are in parentheses*

\*\*\*  $p < .01$ , \*\*  $p < .05$ , \*  $p < .1$

*Source: author's elaboration.*

Table 6.5. presents the results using different subsamples based on the specification in column (2) of Table 6.4, which includes only real GDP per capita and population growth as controls. Column (1) restricts the analysis to high-income countries using the World Bank's classification (for more details on country samples see Annex 11.2). Column (2) estimates the model using the full countries sample from 2010 onward. As previously discussed, the financial landscape changed significantly following the GFC, particularly due to increased regulation. Although the crisis began in 2007, its most pronounced effects were felt in 2008 and 2009. Finally, column (3) combines both criteria, estimating the model for high-income countries for the post-2010 period.

In all three cases, the linear coefficient of private credit is negative and statistically significant. The quadratic term is not statistically significant in the first two specifications, indicating no strong evidence of a non-linear relationship in these cases. In contrast, in column (3), the quadratic term becomes positive and statistically significant, suggesting that the negative effect of private credit may diminish or even reverse at higher levels of financial development within this specific context.

**Table 6.5.** Regression results: GDP per capita growth and private credit by banks (5-year averages) using reduced samples

|                                    | (1)                     | (2)                   | (3)                       |
|------------------------------------|-------------------------|-----------------------|---------------------------|
| Priv. Credit by Banks (5y)         | -.03434**<br>(.01645)   | -.09436**<br>(.03902) | -.16453***<br>(.03607)    |
| Squared Priv. Credit by Banks (5y) | .00007<br>(.00006)      | .00024<br>(.00016)    | .00037***<br>(.00012)     |
| Log Real GDP pc (US\$)             | -2.16714***<br>(.65494) | -.27275<br>(5.12024)  | -10.08434***<br>(2.50182) |
| Pop. Growth (5y)                   | -.38395*<br>(.21127)    | -.67568<br>(.52812)   | -.54604**<br>(.23831)     |
| Observations                       | 589                     | 485                   | 160                       |
| R-squared                          | .43575                  | .53219                | .70621                    |
| Country FE                         | YES                     | YES                   | YES                       |
| Year FE                            | YES                     | YES                   | YES                       |

*Standard errors are in parentheses*

\*\*\*  $p < .01$ , \*\*  $p < .05$ , \*  $p < .1$

*Source: author's elaboration.*

Table 6.6. presents the results for the full sample using the share of employment in the financial sector as the dependent variable. It is worth noting that the dependent variable in this case is only available for certain countries and from 1970 only, which reduces sample size. In columns (1) through (3), both  $b_1$  and  $b_2$  are statistically insignificant. However, in column (4), when controlling for regulation quality (and therefore reducing the sample size), the coefficient on the linear term  $b_1$  is negative and significant, while the coefficient on the quadratic term  $b_2$  is positive and significant. This result, along with column (3) of Table 6.5, suggests a U-shaped relationship that contrasts with the inverted U-shape proposed by Cecchetti and Kharroubi (2012).

**Table 6.6.** Regression results: GDP per capita growth and share of employment in the financial sector (5-year averages) using full sample

|                                 | (1)                       | (2)                       | (3)                       | (4)                         |
|---------------------------------|---------------------------|---------------------------|---------------------------|-----------------------------|
| Finance Employment (5y)         | 15.26932<br>(151.93487)   | 17.33481<br>(152.81116)   | 9.0553<br>(147.99953)     | -238.42529**<br>(105.28374) |
| Squared Finance Employment (5y) | -590.83897<br>(2322.8235) | -519.76924<br>(2329.5838) | -482.15654<br>(2302.2548) | 4056.9674***<br>(1353.8552) |
| Log Real GDP pc (US\$)          | -3.58377***<br>(1.08192)  | -3.55225***<br>(1.04995)  | -3.55168***<br>(1.08497)  | -4.46449**<br>(2.01176)     |
| Pop. Growth (5y)                |                           | .1311<br>(.24554)         | .13586<br>(.24902)        | .16771<br>(.30605)          |
| CPI Inflation (5y)              |                           |                           | -.03591<br>(.0372)        | -.0026<br>(.00956)          |
| Regulatory quality              |                           |                           |                           | -.05421<br>(.03872)         |
| Observations                    | 181                       | 181                       | 181                       | 119                         |
| R-squared                       | .62969                    | .63048                    | .63553                    | .72956                      |
| Country FE                      | YES                       | YES                       | YES                       | YES                         |
| Year FE                         | YES                       | YES                       | YES                       | YES                         |

*Standard errors are in parentheses*

\*\*\*  $p < .01$ , \*\*  $p < .05$ , \*  $p < .1$

*Source: author's elaboration.*

Table 6.7. presents the results for financial sector employment using the same sample selections as Table 6.5. In column (1), the linear and the quadratic coefficients are statically insignificant. However, in columns (2) and (3), both coefficients become statistically significant and form a parabolic relationship, highlighting, once more, that the results are not robust across different sample specifications.

**Table 6.7.** Regression results: GDP per capita growth and share of employment in the financial sector (5-year averages) using reduced samples

|                                 | (1)                       | (2)                        | (3)                          |
|---------------------------------|---------------------------|----------------------------|------------------------------|
| Finance Employment (5y)         | 22.66303<br>(174.00468)   | -393.31943*<br>(225.32848) | -539.12405***<br>(175.10141) |
| Squared Finance Employment (5y) | -615.97568<br>(2509.0176) | 5127.7028*<br>(2507.743)   | 6953.0885***<br>(1879.0462)  |
| Log Real GDP pc (US\$)          | -4.28391***<br>(.98993)   | -8.05682***<br>(2.47815)   | -8.46733**<br>(3.31827)      |
| Pop. Growth (5y)                | .04058<br>(.26384)        | -.2773<br>(.34772)         | -.18326<br>(.30592)          |
| Observations                    | 167                       | 73                         | 65                           |
| R-squared                       | .65565                    | .68678                     | .68349                       |
| Country FE                      | YES                       | YES                        | YES                          |
| Year FE                         | YES                       | YES                        | YES                          |

*Standard errors are in parentheses*

\*\*\*  $p < .01$ , \*\*  $p < .05$ , \*  $p < .1$

*Source: author's elaboration.*

All results remain robust when using GDP growth instead of GDP per capita growth as an alternative measure of economic performance. Although the results in this section provide a more refined approach compared to the previous section and previous work by Cecchetti & Kharroubi (2012), drawing definitive conclusions remains difficult. The relationship between financial development and GDP per capita growth fluctuates across different specifications, and the assumption of a quadratic form may impose an artificial structure on the data. The lack of consistency in results suggests that the connection between financial development and growth is not straightforward.

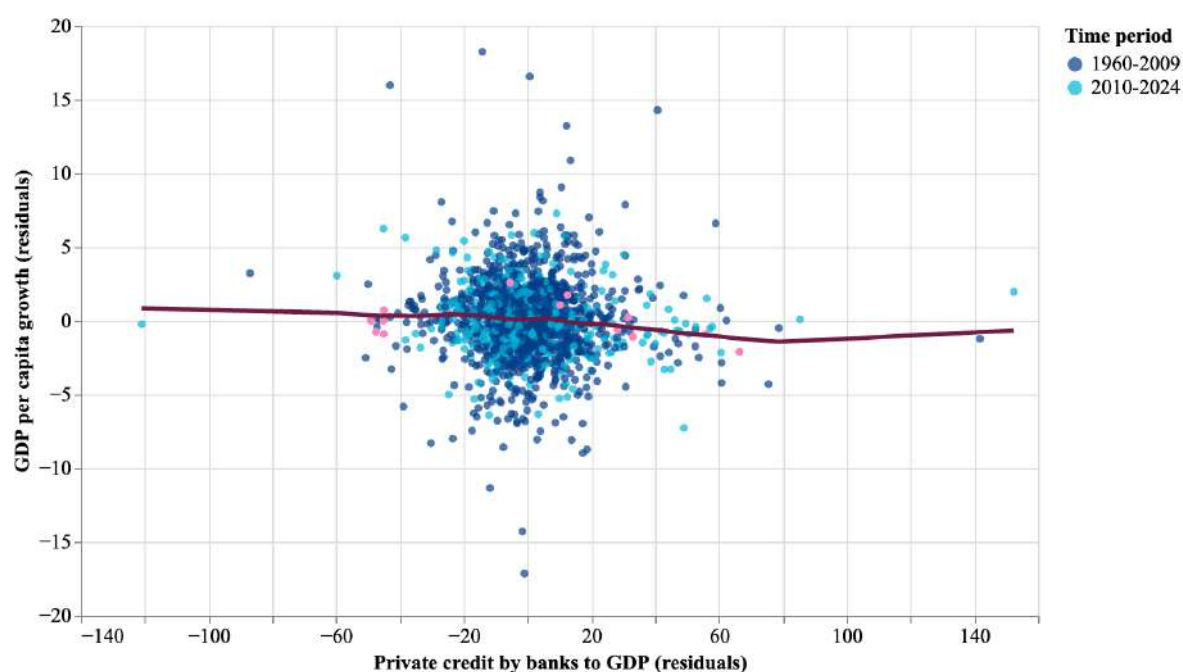
The figures below present the relationship between GDP per capita growth and two measures of financial development: private credit by banks (Figure 6.1) and financial sector employment (Figure 6.2), using a LOWESS (locally weighted scatterplot smoothing) method applied to five-year averaged residuals, which control for GDP, population growth and, more importantly, fixed effects (country and year). This nonparametric approach allows for the visualisation of potential nonlinear relationships without imposing a specific functional form.

As data points are widely dispersed without a discernible trend, making the LOWESS curves appear relatively flat, both graphs show no strong systematic relationship between financial

development and GDP growth once other confounding factors have been accounted for. Additionally, the colours are evenly distributed, indicating no evident pattern distinguishing pre and post-GFC periods.

The LOWESS method is particularly useful in this context because it smooths local variations while avoiding rigid functional assumptions, such as the quadratic forms imposed in previous regressions. The results suggest that once accounting for country and time-specific factors and other variables such as GDP per capita levels and population growth, financial development, at least as measured by private credit and employment in finance, does not display a robust, systematic association with economic growth.

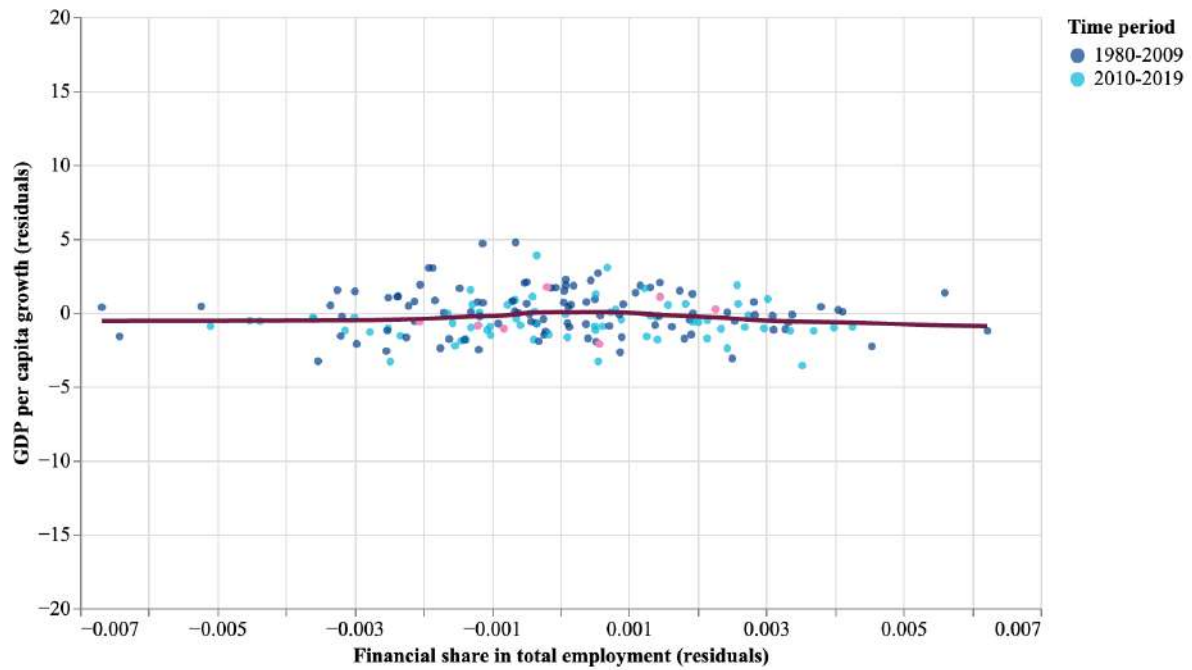
**Figure 6.1.** Private credit by banks to GDP ratio vs GDP per capita growth (five-year averaged residuals)



*Note: observations for UK highlighted in pink, LOWESS curve used to capture non-linearities.*

*Source: authors' elaboration.*

**Figure 6.2.** Financial share in total employment ratio vs GDP per capita growth (five-year averaged residuals)



*Note: observations for UK highlighted in pink, LOWESS curve used to capture non-linearities.*

*Source: authors' elaboration.*

## 7. Discussion of results

### 7.1. All results prove correlation rather than causality

First, this study finds that the methodological design in the existing literature that employs a second-degree polynomial to model the relationship between financial development and economic performance suffers from several critical flaws.

The most significant issue is its failure to establish a causal link between the two variables. The literature asserts that beyond a certain threshold, the financial sector becomes detrimental to economic growth. However, this conclusion is flawed, as it assumes causality without providing robust empirical evidence. The model is plagued by endogeneity, with multiple confounding factors potentially influencing both variables simultaneously. Omitted variables could be driving the observed relationship, or there could also exist a reverse causal link if economic growth itself influences financial sector size.

These methodological weaknesses potentially introduce bias into the results, making any inferred relationship correlational rather than causal. Consequently, the findings should be interpreted with caution, as they do not provide definitive evidence of a direct causal effect.

Moreover, Cecchetti & Kharroubi (2012) make use of a graph to defend their results, using demeaned financial development on the horizontal axis and demeaned GDP per capita growth on the vertical axis. However, by doing so they are not controlling for country or year-specific characteristics (fixed effects) or control variables, mirroring the previously discussed limitations (see Section 5.6). To account for this issue and provide a better graphical representation of the relationship between finance and growth, this study plots the residuals of the running variables. These correlations control for GDP, GDP per capita, population growth, and more importantly, country and year-fixed effects.

### 7.2. There is no clear correlation between financial development and growth

By plotting the results from the first five columns of Tables 6.1 to 6.7 in Equation 2, this section examines whether there exists an inverted parabola shape, and therefore, a growth maximising level of private credit and financial employment (see Table 7.1).

**Table 7.1. Authors' regression results coefficients compared to the academic literature**

| Independent variable    | Paper                           | $b_1$               | $b_2$               | Maximizing point |
|-------------------------|---------------------------------|---------------------|---------------------|------------------|
| Private credit          | Arcand et al., 2012             | 5.262***            | -2.633**            | 99.9%            |
|                         | Cecchetti & Kharroubi, 2012     | 0.036***            | -0.018***           | 100%             |
|                         | Authors (C&K 2012 replicate)    | 0.05511***          | -0.00028***         | 98.4%            |
|                         | <b>Authors (improved model)</b> | <b>0.00579</b>      | <b>-0.00011</b>     | <b>N/A</b>       |
| Private credit by banks | Cecchetti & Kharroubi, 2012     | 0.0369***           | -0.0196***          | 94.1%            |
|                         | Authors (C&K 2012 replicate)    | 0.03946***          | -0.0002***          | 98.7%            |
|                         | <b>Authors (improved model)</b> | <b>-0.05052**</b>   | <b>0.00011</b>      | <b>N/A</b>       |
| Employment              | Cecchetti & Kharroubi, 2012     | 3.345***            | -43.35**            | 3.86%            |
|                         | Authors (C&K 2012 replicate)    | 159.3129***         | -1769.859***        | 4.50%            |
|                         | <b>Authors (improved model)</b> | <b>-238.42529**</b> | <b>4056.9674***</b> | <b>N/A</b>       |

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

Note: 'Authors (improved)' refers to the suggested corrected model, with full country and year sample, clustered standard error, fixed effects, and a relevant set of controls. Only statistically significant maximizing points are presented in this table.

Source: author's elaboration.

First, when replicating the methodological design of Cecchetti and Kharroubi (2012) (ignoring its previously discussed methodological flaws), this study finds that the growth-maximising level of private credit is 98.4% of GDP and the growth-maximising level of private credit by banks of 98.7% of GDP (Table 7.1). These are in line with the findings in the literature, where ratios range from 94.1% to 100% (Arcand et al., 2012; Cecchetti & Kharroubi, 2012). Based on these values, the UK crossed that threshold around 1996 and has stayed above it ever since. It reached a peak of 192% in 2009 and currently stands just below 120% (see Figure 11.1 in Annex 11.3).

However, even if one were to accept the methodological approach and the arbitrary sample selection of the replication exercise, the results show a lack of robustness when fixed effects are incorporated, indicating their fragility. This remains true also when clustering standard errors, testing different sample sizes, and incorporating alternative control variables.

In terms of financial sector's share in total employment, the literature identifies a turning point around 3.9%, while the replication in this study finds a turning point of 4.5%. As shown in Figure 4.7, the UK's current share of financial employment stands at 3.2%. Nevertheless, after refining the methodological design – clustering standard errors, incorporating time and country fixed effects, and testing different samples and control variables – the estimated coefficients

retain the opposite signs as those found by Cecchetti & Karroubi (2012). Therefore, there is no evidence a growth-maximising level of financial employment in the economy, as estimates point to a global minimum (Table 7.1).

As such, the lack of robustness of these results when including fixed effects and addressing some methodological flaws indicates their fragility. No meaningful relationship can thus be determined between the different metrics of financial development and growth, suggesting the unreliability of previous findings.

Adding to the lack of robustness in the regression estimates, the graphical analysis provides further evidence against the existence of a systematic relationship between financial development and GDP per capita growth (Figures 6.1-6.2). The LOWESS curves applied to the residuals of financial development reveal no clear trend, reinforcing the conclusion that any observed inverted parabolic pattern in the regressions is likely an artefact of model specification rather than a true economic relationship. By allowing for flexible, nonparametric visualisation without imposing a predefined functional form, these graphs serve as a powerful tool to discard spurious patterns and highlight the absence of a meaningful relationship.

Taken together, both the inconclusive regression results and graphical evidence strongly suggest that it is not possible to claim an inverted U-shaped relationship between financial development and economic growth.

### **7.3. Results remain unchanged when focusing solely on high-income countries**

As previously mentioned, this study finds no clear evidence that financial sector development either supports or hinders economic growth beyond a certain threshold on a global scale. Restricting the sample to high-income countries could provide a more relevant comparison for the UK. However, no clear inverted U-shape emerges from this sample. The regression analysis yields statistically insignificant results. If anything, plotting residuals from the data suggests a U-shaped relationship, indicating that financial development may initially hinder growth before supporting it beyond a certain point. However, this tipping point is critically dependent on a limited number of outliers on the right end of the distribution (see Figures 11.2-11.3 in Annex 11.3).

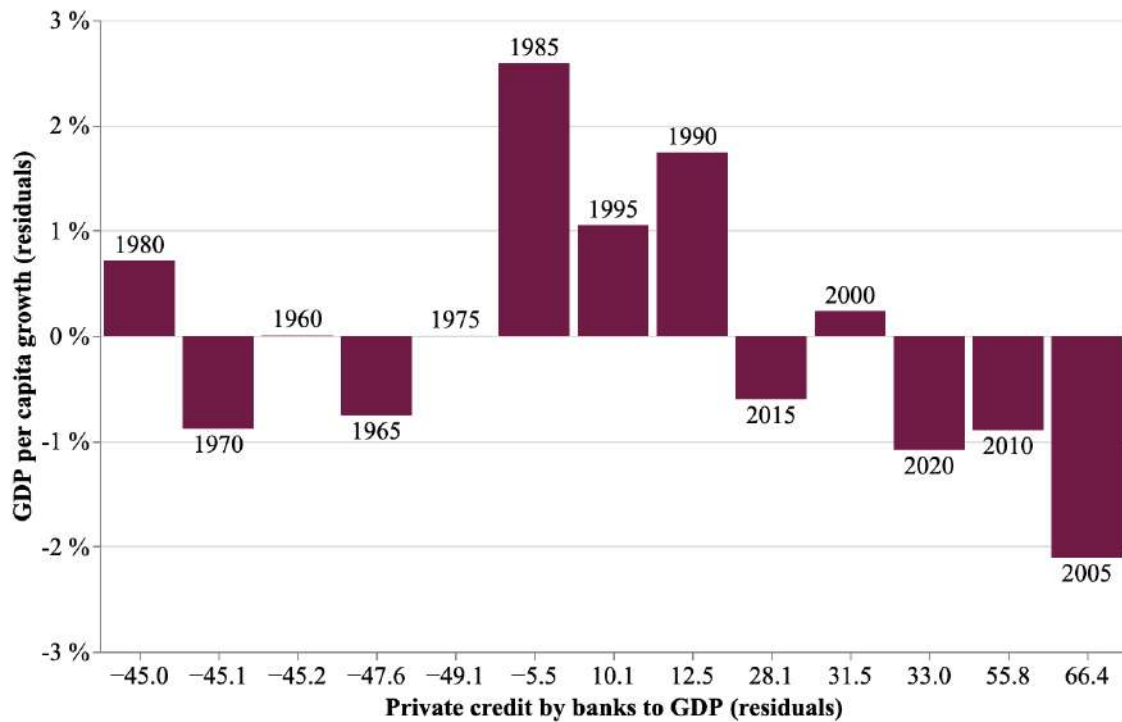
#### **7.4. Is Britain's financial services sector too large? The importance of time-specific characteristics**

Zooming in on the residuals for UK observations reveals a somewhat different pattern. Interestingly, the inverted U-shaped relationship described in the literature appears to fit UK private credit residuals well, suggesting that financial development has positively contributed to UK growth up to a threshold (Figure 7.1). However, no relevant relationship can be advanced for financial employment residuals (Figure 7.2). It is important to note that these figures represent regression residuals, not absolute values. Residuals capture the portion of growth that is not explained by country and year-specific characteristics or overall economic development and population growth.

There are however two major caveats. First, this relationship is purely correlative, as nothing causal can be advanced in this study. Second, while higher financial development correlates with negative growth rates, this finding is strongly linked to time, as increased financial development correlates with negative GDP growth from 2005 onwards (Figure 7.1). In this respect, it may very well be true that increased financial development has entailed lower GDP per capita growth in Britain, but this is likely due to time-specific factors unique to the UK, rather than the size of the financial sector itself.

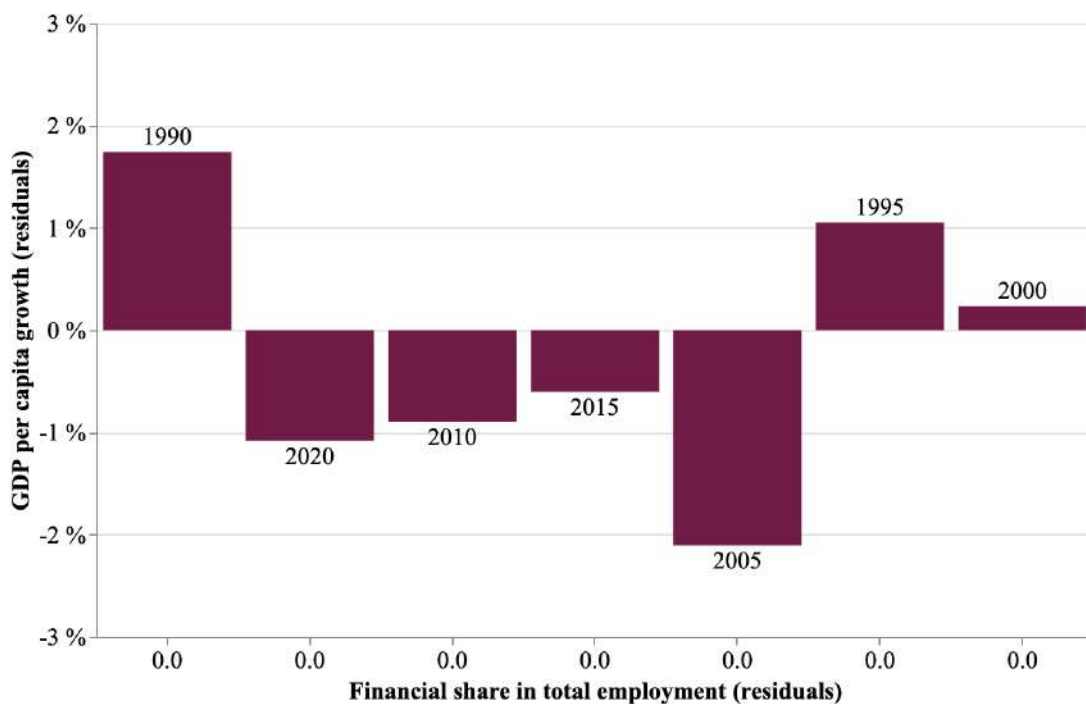
As such, linking back to Sections 1 through 4 of this report, this study suggests that the perceived negative effect of financial development on growth in Britain is likely due to the long-run effects of the GFC, durably inhibiting financial sector productivity growth, rather than financial development per se. Any policy conclusions to UK policymakers derived from this study should unmistakably devote a large role to time-specific and context-specific factors rather than oversimplified generalist conclusions on financial sector size.

**Figure 7.1.** Private credit by banks to GDP ratio vs GDP per capita growth, United Kingdom (five-year averaged residuals)



Source: authors' elaboration.

**Figure 7.2.** Financial share in total employment ratio vs GDP per capita growth, United Kingdom (five-year averaged residuals)



Source: authors' elaboration.

## 8. Policy recommendations

The evidence from this study does not support the hypothesis of an inverted U-shaped relationship between financial development and economic performance, suggesting that the financial sector is not excessively large in the UK economy, contrary to claims made by some authors. Accordingly, the recommendations that follow focus on enhancing the financial sector's positive contribution to the UK economy, rather than seeking to reduce its size through regulation or exposing it to greater risks through poor deregulation.

### 8.1. Support UK small businesses

Small and medium-sized enterprises (SMEs) represent a significant share of employment, innovation, and regional development, and their ability to grow and invest depends heavily on a supportive financial environment. Ensuring that SMEs have consistent, affordable access to finance is vital, not only to stimulate entrepreneurship and job creation, but also to contribute to broader economic growth, competitiveness, and productivity across sectors. Evidence from the literature further supports this view, showing that credit directed toward enterprises, rather than households, is more strongly associated with long-term growth and stability (Beck et al., 2012, 2014).

This proposal outlines a targeted strategy to strengthen SME access to finance to unlock their full potential and sustain their role in the UK economy.

The first key recommendation is to expand the Growth Guarantee Scheme (GGS). Introduced in June 2024 as a successor to earlier COVID-19 support programmes, the GGS plays a vital role in facilitating SME lending by sharing risk between lenders and the government. As of September 2024, 82% of the funds have supported small businesses located outside London, and 84% have gone to firms with fewer than 50 employees, underscoring the scheme's strong regional and small-business focus (British Business Bank, 2024). In particular, the Harvard Kennedy School has identified access to finance as a key constraint to growth in UK regions beyond London and the South East (Stansbury et al., 2023). The scheme has been predominant in sectors such as manufacturing, construction, retail, and professional services, where access to affordable finance is essential for investment and growth. With a current loan cap of £2 million and a maximum loan term of six years, the scheme could be made more effective by

increasing the loan ceiling, extending repayment periods, and expanding its overall budget. Given that GGS is currently due to expire in March 2026, contingent on the government's fiscal headroom and willingness to underwrite additional liabilities, consideration should also be given to regularising the scheme as a permanent mechanism to provide SMEs with consistent access to finance. These enhancements would give SMEs greater flexibility to pursue long-term investments and improve access to finance.

A second component involves addressing the use of personal guarantees. While these are commonly required by lenders as a form of security, especially when dealing with small businesses that may lack significant collateral or credit history, there is growing concern that this practice may be preventing access to credit. The Federation of Small Businesses issued a super-complaint on this issue, which led the FCA in March 2024 to launch an investigation into the use of personal guarantees in loans to sole traders and small partnerships borrowing under £25,000 (FCA, 2024). Although the FCA's remit does not extend to lending to limited companies, its initiative to collect data, collaborate with the Financial Ombudsman Service, and consider regulatory guidance is a step in the right direction. Reducing over-reliance on personal guarantees can help foster a more inclusive and enabling lending environment for small businesses.

Furthermore, a key principle should be to avoid expanding regulatory restrictions that could involuntarily constrain lending, as these often add compliance burdens without a clear improvement in outcomes for businesses or lenders. A more focused, proportionate regulatory approach would help maintain a healthy flow of credit while ensuring appropriate protections remain in place.

Together, these measures aim to remove key barriers to finance, enhance the resilience of the SMEs, and ensure that small businesses continue to play a leading role in driving innovation, employment, and economic dynamism across the UK.

## **8.2. Strengthen regulatory coordination**

This study challenges the notion that the UK's financial sector has grown beyond its optimal size, suggesting that the question facing policymakers is not whether to shrink the sector, but how to shape it to better supports the wider economy. One area where institutional design can

have a significant impact is a more streamlined financial regulatory framework. The current regulatory architecture in the UK, with responsibilities divided across multiple bodies, introduces complexity that can dilute oversight and reduce responsiveness.

The UK's financial regulatory system has evolved into a complex framework involving multiple public bodies, each with important and distinct responsibilities. The Prudential Regulation Authority (PRA), within the Bank of England (BoE), supervises banks, insurers, and investment firms' financial soundness. The Financial Conduct Authority (FCA) regulates the conduct of financial services, focusing on protecting consumers, promoting competition, and ensuring market integrity. The Payment Systems Regulator (PSR) oversees payment systems to promote fair access and innovation. The Financial Ombudsman Service (FOS) provides independent dispute resolution between consumers and financial firms. Finally, the Competition and Markets Authority (CMA) oversees competition across sectors, including financial services.

However, institutional fragmentation, especially in high-risk investments, payments, and consumer protection, has created regulatory grey zones, coordination costs, and, in some cases, harmful supervision gaps. The London Capital & Finance (LCF) case starkly illustrates the risks of these institutional divides, where a lack of regulatory clarity contributed to significant investor losses and delayed intervention (Serious Fraud Office, 2024). The government's recent decision to integrate the PSR into the FCA is a response to this fragmentation, reducing mandate duplications (Prime Minister's Office, 2025).

Yet, further consolidation, particularly of diverse and complex responsibilities, should proceed with caution. Experts have raised concerns that the FCA's rapidly expanding remit could strain its capacity and undermine its operational effectiveness, especially following the transfer of consumer credit regulation from the Office of Fair Trading (Gloster, 2020). The investigation into the LCF case recommended that HM Treasury reconsider the optimal scope of the FCA's responsibilities, noting that comparable jurisdictions often divide such roles across multiple bodies (Gloster, 2020). These findings suggest that instead of pursuing further institutional mergers, reform efforts should focus on strengthening coordination amongst existing regulators.

Improving regulatory coherence through structured inter-agency mechanisms would allow for better oversight without overburdening a single institution. This could include updated memoranda of understanding (MoUs) between key regulators, such as the agreement between the FCA and the BoE, which is now subject to annual review. In addition, more consistent joint supervision and enforcement initiatives, shared data systems for risk monitoring, and even the creation of a formal coordination council (chaired by HM Treasury or the BoE) would help align regulatory strategies while maintaining clarity of roles.

This approach would allow the UK to preserve the sectoral expertise of different institutions while addressing fragmentation in oversight and accountability. It also avoids the risk of overstretching any single entity. A more coordinated architecture, rather than a more centralised one, is better suited to manage the increasing complexity of financial innovation and ensure that regulation remains both effective and adaptable.

### **8.3. Foster regional investment**

This report has shown that London captures a predominant share of the UK's FSS value added. While this concentration has supported growth in the capital, it has limited the sector's ability to act as a broader engine of economic development. Addressing this could unlock the FSS' full potential as a driver of inclusive and sustainable growth across the UK.

To support more balanced growth, policy could aim to foster a more attractive and competitive investment environment across other UK regions. Investment decisions, particularly from financial institutions and private capital, are typically guided by the potential for strong returns, which highlights the relevance of addressing structural factors that currently limit the appeal of areas beyond London. Reducing these constraints may help broaden the geographic reach of financial sector investment and strengthen its contribution to regional economies.

One critical limitation is underinvestment in transport infrastructure. Major cities such as Birmingham and Manchester continue to lag behind comparable European urban centres in terms of connectivity, which undermines their economic performance and limits integration with national and international markets (Shehadi, 2023; Stansbury et al., 2023). Improving regional transport infrastructure would not only enhance productivity and mobility but also improve the business case for financial investment in these areas.

A complementary strategy is to encourage investment in R&D beyond London and the South East, where it remains heavily concentrated. This is particularly relevant in the context of FinTech, which is a major driver of financial sector investment (Farraj, 2024). Supporting the growth of FinTech and other high-value technology sectors in other regions could attract new capital flows and foster innovation ecosystems that are currently too centralised.

Together, these initiatives would help unlock the financial services sector's contribution to regional growth by creating the conditions necessary for private investment to flow more evenly across the UK. While such efforts would require sustained public investment and coordination, they represent a strategic path toward reducing regional disparities and increasing the financial sector's value to the broader economy. They also align closely with the government's ongoing devolution agenda which seeks to empower local authorities and promote regionally led growth through deeper devolution deals and targeted investment (McMahon, 2024). Strengthening the role of the financial sector in this context could amplify the impact of devolution by broadening access to capital and financial expertise across the country.

## 9. Conclusion

The main objective of this study is to challenge the hypothesis that the UK's financial sector has grown beyond its optimal size. To do so, it builds on the existing literature that finds an inverted U-shaped relationship between financial development and growth. By replicating this methodology and then enhancing the design to account for severe methodological flaws, this study concludes that it is not possible to claim an inverted U-shaped relationship between financial development and growth, as financial development does not display a robust, statistically significant, and systematically relevant relationship with growth. As such, this study crucially argues that any identification strategy finding that the FSS is too large cannot prove anything causal, as endogeneity in the causal relationship between financial development and growth invalidates any causal identification.

However, correlative findings from this research do point to one important finding that should serve as a basis to orientate UK financial policymaking. It appears that Britain may have actually had a growth-maximising level of financial development preceding 2005, but this is most likely due to time-specific characteristics (the GFC) rather than financial development in itself. Accordingly, time and context-specific factors should play a large role in the formulation of any policy conclusions from this field of study.

As such, the policy recommendations presented in this report focus on enhancing the UK's financial sector's ability to support broad-based economic growth. First, targeted measures to improve SME access to finance, such as strengthening the Growth Guarantee Scheme and reducing reliance on personal guarantees, could increase investment, employment, and innovation among small businesses across the country. Second, instead of further consolidating regulatory bodies, efforts to improve coordination between existing institutions, through clearer mandates, joint supervision, and shared data infrastructure, could make regulation more effective without overstressing individual regulators. Third, creating a more attractive investment environment in regions outside London, particularly through improved infrastructure and support for R&D, could help spread the benefits of financial sector development more evenly and unlock its potential as a catalyst for regional growth.

Taken together, the findings from this study suggest that debates over the absolute size of UK financial services may be less relevant than the question of how the FSS functions and where its resources are directed. Rather than focusing on limiting financial development, the emphasis should shift toward designing institutions and policies that better align the sector with productive investment, innovation, and inclusive growth. By grounding policymaking in evidence, context, and institutional design, rather than thinking of abstract thresholds, this approach offers a more constructive path forward for leveraging finance in support of the UK's long-term economic objectives.

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## 11. Annexes

### Annex 11.1. Data collection description

#### 11.1.1 World Bank Financial Structure Database (FSD) and Global Financial Development Database (GFDD)

The Financial Structure and Global Financial Development Databases from the World Bank were last updated in 2019 and 2022 respectively, and offer a homogenous cross-country mapping of various financial sector statistics (World Bank, 2019, 2022). These databases disclose data on financial institutions and markets relative to: financial depth (i.e. size), financial access (i.e. ease of use), financial efficiency (in providing financial services) and financial stability (Beck et al., 2000; Čihák et al., 2012). From this dataset, the following variables were extracted:

**Table 11.1.** World Bank FSD and GFDD data collection description

| Variable                                                                    | Calculation method                                                                                                    | Source                                                                                                                 |
|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Country                                                                     |                                                                                                                       |                                                                                                                        |
| Country code                                                                |                                                                                                                       |                                                                                                                        |
| Year                                                                        |                                                                                                                       |                                                                                                                        |
| Private credit by deposit money banks to GDP (%)                            | $0.5 \times \frac{\left[ \frac{F_t}{P_{et}} + \frac{F_{t-1}}{P_{et-1}} \right]}{\left[ \frac{GDP_t}{P_{at}} \right]}$ | IMF International Financial Statistics, 2019<br>(private credit, GDP in local currency, end-of-period CPI, annual CPI) |
| Private credit by deposit money and other financial institutions to GDP (%) | where $F$ is credit to the private sector, $P_e$ is end-of period CPI, and $P_a$ is average annual CPI                |                                                                                                                        |

*Source: authors' elaboration.*

#### 11.1.2 International Labour Organisation SDG Labour Market Indicators Database (ILOSDG)

The ILO's SDG Labour Market Indicators Database reports 14 Sustainable Development Goals (SDG) indicators to the United Nations (ILO, 2025). For this analysis, data was extracted from Goal 8 ('Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all') under Target 8.2 ('Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including

through a focus on high value-added and labour-intensive sectors’) (UN Statistics, 2018). From this dataset, the following variables were extracted:

| Table 11.2. International Labour Organisation SDG data collection description                     |                                                                                                                                                                                               |                                                            |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Variable                                                                                          | Calculation method                                                                                                                                                                            | Source                                                     |
| Country                                                                                           |                                                                                                                                                                                               |                                                            |
| Year                                                                                              |                                                                                                                                                                                               |                                                            |
| Output per worker (GDP constant 2021 international \$ at PPP) - ILO modelled estimates, Nov. 2024 | $LP_t = \frac{GDP_t}{Total\ employed\ persons}$ <p>where <math>LP_t</math> is the real GDP per person employed for year <math>t</math></p>                                                    | ILO Modelled Estimates (ILOEST)                            |
| Annual growth rate of output per worker (GDP constant 2021, international \$ at PPP) (%)          | $GEP_t^{GDP} = \frac{LP_t - LP_{t-1}}{LP_{t-1}} \times 100$ <p>where <math>LP_{t-1}</math> is the real GDP per person employed for year <math>t - 1</math> (at constant base year prices)</p> | ILO Labour Market-Related SDG Indicators (ILOSDG database) |

Source: authors' elaboration.

### 11.1.3. OECD Structural Analysis Industry Database (STAN)

The OECD's STAN Database for Structural Analysis is based on member countries' annual national accounts, under 2008 System of National Accounts recommendations (OECD, 2025b). The database discloses annual measures of output, value added, labour input, investment and capital stock, for each member country (Horvát & Webb, 2020). From this dataset, the following variables were extracted:

| Table 11.3. OECD STAN data collection description   |                                                      |                                                          |
|-----------------------------------------------------|------------------------------------------------------|----------------------------------------------------------|
| Variable                                            | Calculation method                                   | Source                                                   |
| Country                                             |                                                      |                                                          |
| Year                                                |                                                      |                                                          |
| Financial and insurance activities total employment | Number of jobs (absolute values, quarterly averages) | OECD STAN Database for Structural Analysis, 2025 edition |
| All activities total employment                     |                                                      |                                                          |

Source: authors' elaboration.

Once these data collected, the following variable was constructed:

**Table 11.4. OECD STAN data construction description**

| Variable                                                            | Calculation method                                                             | Source                                                   |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------|
| Share of financial and insurance employment in total employment (%) | $L_{FSS} = \frac{FSS \text{ employment}}{Total \text{ employment}} \times 100$ | OECD STAN Database for Structural Analysis, 2025 edition |

*Source: authors' elaboration.*

#### 11.1.4. World Bank World Development Indicators (WDI)

The World Bank's World Development Indicators (World Bank, 2025a) presents global development data that used in the regression to control for country-specific and time-specific fixed effects. The control variables extracted from this dataset are the following:

**Table 11.5. World Bank WDI data collection description**

| Variable                                                          | Calculation method                                                                                                                                                                                                                               | Source                                                                          |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Country                                                           |                                                                                                                                                                                                                                                  |                                                                                 |
| Year                                                              |                                                                                                                                                                                                                                                  |                                                                                 |
| GDP per capita (current US\$)                                     | $GDPPC = \frac{GDP}{midyear \text{ population}}$                                                                                                                                                                                                 | World Bank national accounts data, and OECD National Accounts data files        |
| Trade (% GDP)                                                     | $T = \frac{\sum(exports, imports)}{GDP} \times 100$                                                                                                                                                                                              | World Bank national accounts data, and OECD National Accounts data files        |
| Inflation, consumer prices (annual %)                             | $I = \left( \frac{\sum P_{it} Q_{i0}}{\sum P_{i0} Q_{i0}} \right) \times 100$<br>where $P_{i0}$ is the price of commodity $I$ at time 0, $Q_{i0}$ is the quantity of commodity $I$ at time 0, $P_{it}$ is the price of commodity $I$ at time $t$ | International Monetary Fund, International Financial Statistics and data files. |
| General government final consumption expenditure (% GDP)          | $G_c = \frac{\sum gov \text{ consumption expenditure}}{GDP} \times 100$                                                                                                                                                                          | World Bank national accounts data, and OECD National Accounts data files        |
| Working population: population ages 15-64 (% of total population) | $WP = \frac{Total \text{ pop aged } 15 - 64}{Total \text{ pop}} \times 100$                                                                                                                                                                      | United Nations Population Division. World Population Prospects: 2024 Revision   |

*Source: authors' elaboration.*

Using that information, the following variable was constructed:

**Table 11.6. World Bank WDI data creation description**

| Variable                                                | Calculation method                                        | Source                                                                        |
|---------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------|
| Working population growth:<br>population ages 15-64 (%) | $\Delta WP = \frac{WP_t - WP_{t-1}}{WP_{t-1}} \times 100$ | United Nations Population Division. World Population Prospects: 2024 Revision |

*Source: authors' elaboration.*

### 11.1.5. World Bank Worldwide Governance Indicators (WGI)

The Worldwide Governance Indicators (WGI) is a compilation of data that describes perceptions of the quality of governance across countries and over time. The only control variable extracted from this dataset was:

**Table 11.7. World Bank WGI data collection description**

| Variable                            | Calculation method                                                                                                                                                                                                                                                                   | Source                                                                                                                                                                   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Regulatory Quality: Percentile Rank | Percentile rank indicates the country's rank among all countries covered by the aggregate indicator, with 0 corresponding to lowest rank, and 100 to highest rank. Ranks have been adjusted to correct for changes over time in the composition of the countries covered by the WGI. | World Bank WGI are produced by Daniel Kaufmann (Natural Resource Governance Institute and Brookings Institution) and Aart Kraay (World Bank Development Research Group). |

*Source: authors' elaboration.*

## Annex 11.2 Country samples for regressions replicating Cecchetti's methodology

**Table 11.8. Regression analysis country samples**

| Group                     | Countries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group 1<br>(50 countries) | Argentina, Australia, Austria, Bangladesh, Belgium, Brazil, Canada, Chile, China, Colombia, Czech Republic, Denmark, Egypt, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, India, Indonesia, Ireland, Italy, Japan, Korea, Luxembourg, Mexico, Morocco, Netherlands, New Zealand, Nigeria, Norway, Pakistan, Philippines, Poland, Portugal, Russia, Slovakia, Slovenia, South Africa, Spain, Sweden, Switzerland, Thailand, Turkey, United Kingdom, United States, Venezuela, Vietnam |
| Group 2<br>(21 countries) | Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Ireland, Italy, Japan, Korea, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, United States                                                                                                                                                                                                                                                                                         |

*Source: authors' elaboration.*

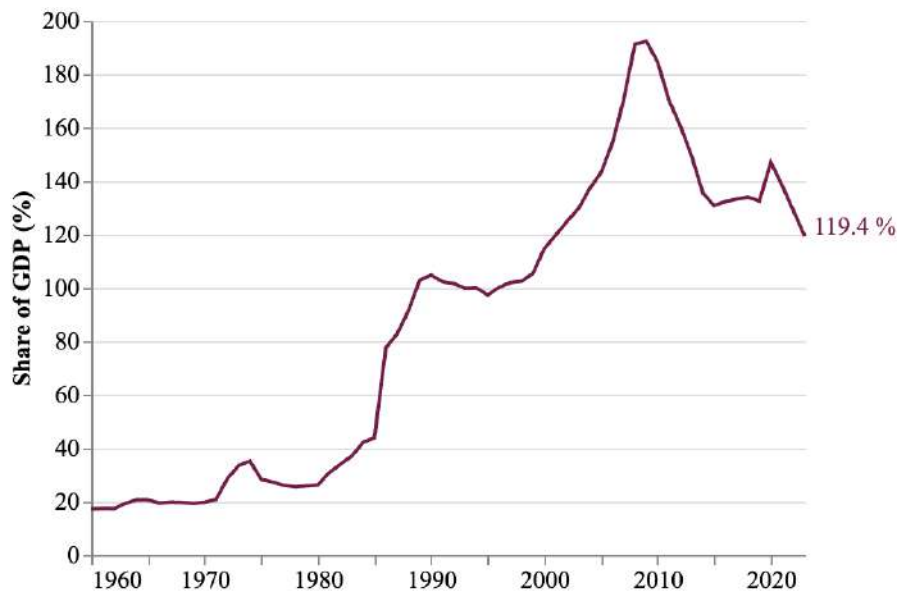
**Table 11.9. High-income country sample**

| Group                         | Countries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| High-income<br>(83 countries) | Andorra, Antigua and Barbuda, Aruba, Australia, Austria, Bahamas, Bahrain, Barbados, Belgium, Bermuda, Brunei Darussalam, Canada, Cayman Islands, Channel Islands, Chile, Croatia, Curaçao, Cyprus, Czech Republic, Czechia, Denmark, Estonia, Faroe Islands, Finland, France, French Polynesia, Germany, Gibraltar, Greece, Greenland, Guam, Hong Kong, Hungary, Iceland, Ireland, Isle of Man, Israel, Italy, Japan, South Korea, Kuwait, Latvia, Liechtenstein, Lithuania, Luxembourg, Macao, Malta, Monaco, Nauru, Netherlands, New Caledonia, New Zealand, Norway, Panama, Poland, Portugal, Puerto Rico, Qatar, Romania, San Marino, Saudi Arabia, Seychelles, Singapore, Sint Maarten, Slovak Republic, Slovenia, Spain, St. Kitts and Nevis, Sweden, Switzerland, Taiwan, China, Trinidad and Tobago, Turks and Caicos Islands, United Arab Emirates, United Kingdom, United States, Uruguay, Virgin Islands |

*Source: authors' elaboration.*

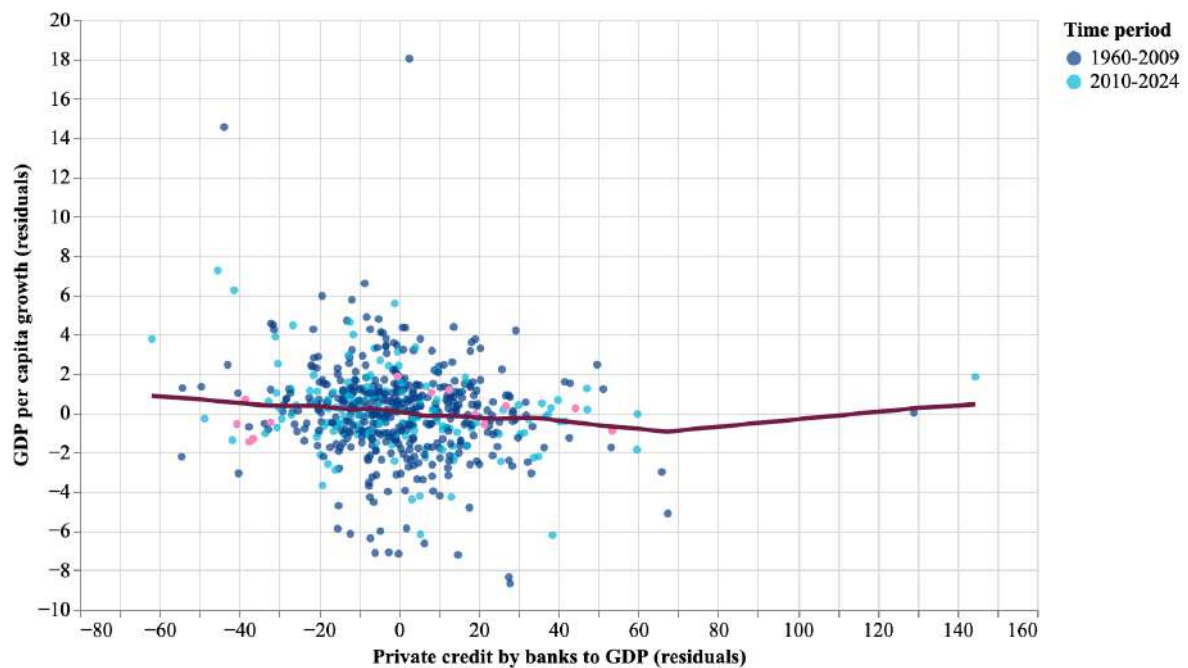
### Annex 11.3. Additional figures

**Figure 11.1.** Private credit to GDP ratio, United Kingdom (1960-2021)



Source: authors' elaboration using World Bank data.

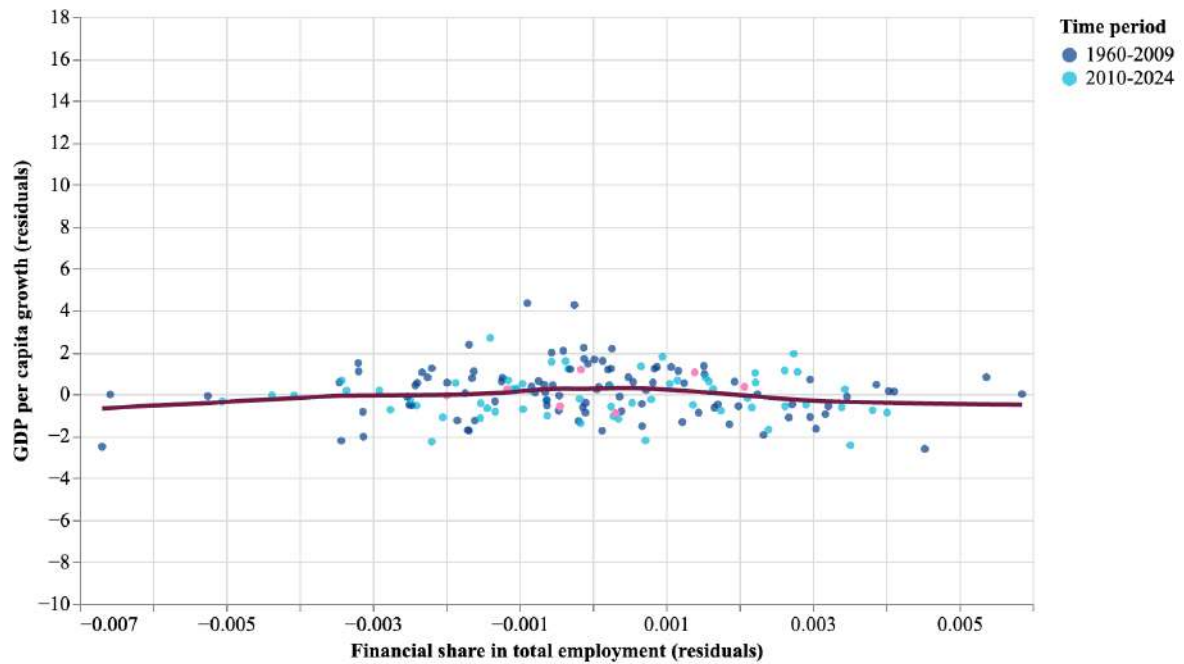
**Figure 11.2.** Private credit by banks to GDP ratio vs GDP per capita growth, high income countries (five-year averaged residuals)



Note: observations for UK highlighted in pink, LOWESS curve used to capture non-linearities

Source: authors' elaboration.

**Figure 11.3.** Financial share in total employment ratio vs GDP per capita growth, high income countries (five-year averaged residuals)



*Note: observations for UK highlighted in pink, LOWESS curve used to capture non-linearities*

*Source: authors' elaboration.*

## Annex 11.4. Final Terms of Reference – March 2025

**Project title:** Is Britain’s Financial Services Sector Too Large?

### Project background and context

The UK financial services sector (FSS) is a cornerstone of the national economy, contributing significantly to GDP, tax revenues, and employment. However, in light of stagnant productivity growth and heightened systemic risks following the Global Financial Crisis (GFC), concerns have emerged over whether further expansion of the sector may hinder, rather than support, broader economic performance.

Christensen et al. (2016) argue that the UK may be experiencing a ‘finance curse’, a phenomenon akin to the Resource Curse, where heavy dependence on the financial sector constrains economic development and produces adverse outcomes such as crowding out, reduced economic diversity, and rent-seeking. Similarly, Baker et al. (2018) suggest that the UK exhibits symptoms of the finance curse, including resource misallocation driven by short-termism and rent-seeking, unintentional brain drain, capital concentration, and Dutch Disease effects.

Concerns about the size of the financial sector and its unintended negative consequences on the economy are not unique to the UK. Despite a rich body of literature documenting a positive relationship between financial sector development and economic growth throughout the 20th century, researchers have increasingly explored the idea that financial development may exhibit diminishing or even negative returns to economic growth beyond a certain threshold. Arcand et al. (2012) and Cecchetti & Kharroubi (2012) suggest an inverted U-shaped relationship between financial sector size and economic performance, with potential risks of resource misallocation and crowding-out effects. In particular, they identify a threshold at which the size of the financial sector begins to negatively affect economic growth: once domestic credit to the private sector reaches 90-100% of GDP (Arcand et al., 2012; Cecchetti & Kharroubi, 2012). However, these claims rely on correlational evidence and face substantial methodological challenges. Empirical research using post-2009 data has also been limited.

Against this backdrop, this study attempts to understand whether the UK’s financial sector has surpassed its optimal size, and how the relationship between finance and growth may be

evolving. This is especially timely, as the Financial Conduct Authority (FCA) was recently granted a secondary objective to support the international competitiveness and growth of the UK economy.

**Primary research question:** Does Britain's financial sector's continued expansion enhance economic productivity, or has it reached a stage where further growth hinders broader economic performance? In other words, is the UK's financial sector too large?

### Research sub-questions

- How can existing research on the finance-growth nexus inform our problem analysis? To what extent does the UK fit into these models, and what are the key limitations or gaps in the existing literature?
- Is there empirical evidence of a non-linear (e.g. inverted U-shaped) relationship between financial sector size and economic growth? And, how can this be empirically measured?
- What are the financial sector's contributions to total output, employment, productivity, investment, tax revenues, and trade?

### Research objectives

- Evaluate the extent to which the financial sector contributes to or constrains the UK's long-term economic growth.
- Test the robustness of the "Too Much Finance" hypothesis using updated and extended datasets. Critically replicate and refine existing econometric methods used in previous literature, particularly the inverted U-shaped relationship proposed by Cecchetti & Kharroubi (2012).
- Evaluate the financial sector's role in the British economy, through investment, productivity, employment, tax revenues, and international trade.
- Provide policy recommendations to enhance the sector's role in supporting broad-based to sustainable and inclusive growth.

### Methodology

This project adopts a mixed-methods approach, combining both qualitative and quantitative analyses. On the qualitative side, it includes a detailed literature review to evaluate the

conceptual and empirical debates surrounding the ‘Too Much Finance’ hypothesis and the broader finance-growth nexus.

For the quantitative component, the study includes an empirical analysis of the UK financial sector’s contribution to output, investment, employment, productivity, trade, and tax revenue. This is followed by panel regression models that replicate and refine Cecchetti & Kharroubi’s (2012) framework, using updated datasets and improved econometric techniques. These include the application of country and time fixed effects, clustered standard errors, and a revised set of control variables. The regression analysis aims to uncover patterns and correlations, while also exploring the extent to which causal relationships can be meaningfully inferred, within the limitations of the data and model design.

### **Data collection**

Data was sourced from:

- World Bank’s Global Financial Development Database (GFDD), World Development Indicators (WDI), and World Governance Indicators (WGI) to construct a panel that spans 1960–2023, with 5-year averages used for key variables.
- OECD’s Structural Analysis Database (STAN)
- UK Office for National Statistics (ONS)
- TheCityUK, HMRC, PwC, and other official publications.

### **Deliverables**

1. A 15,000-word Capstone Report, including an Executive Summary, empirical findings, and policy recommendations.
2. A presentation to the FCA outlining the research insights and proposed policy directions.



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